

# BFS Traversal

Step 1: Start at E

## Legend



Complete



Processing



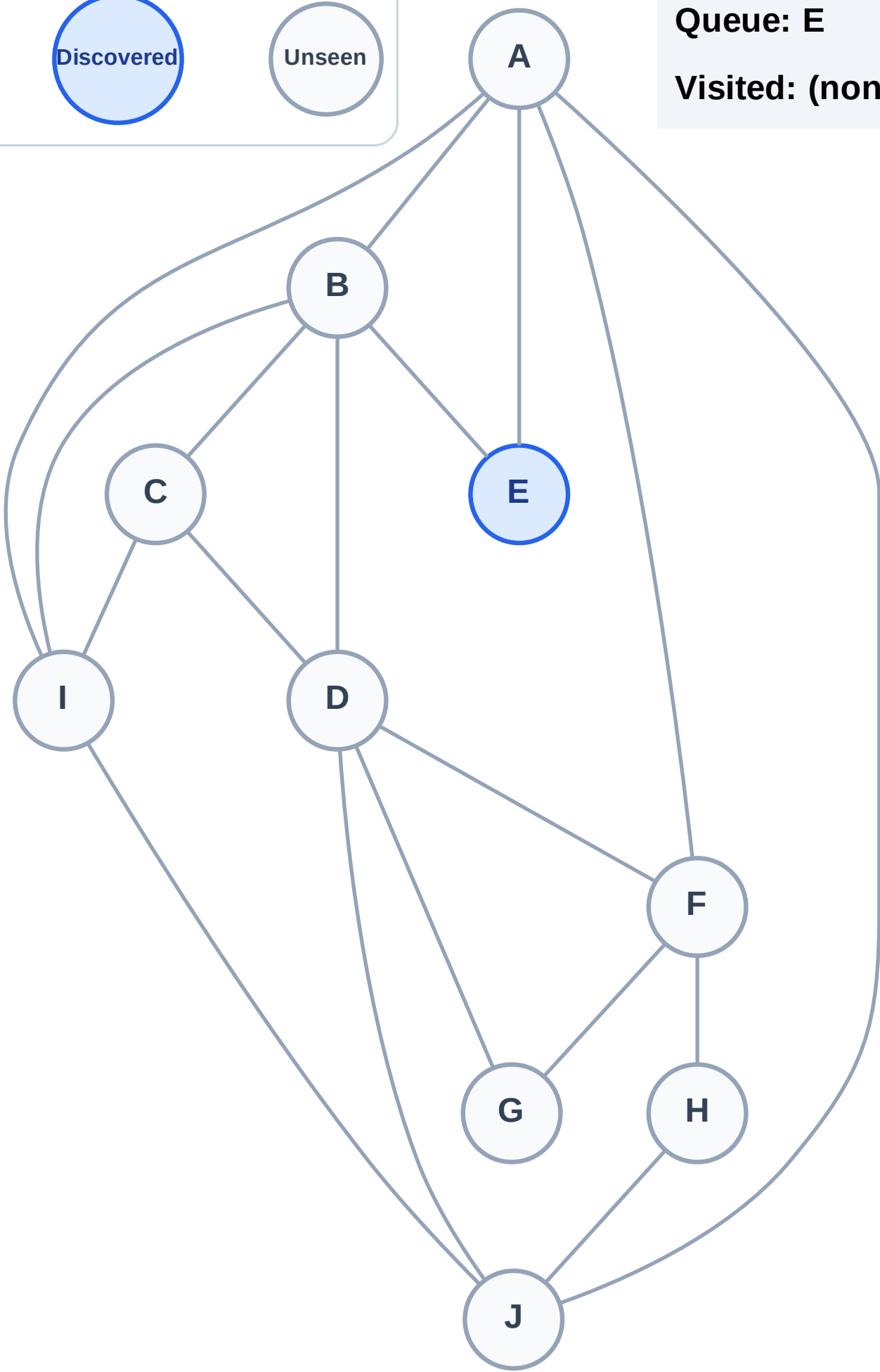
Discovered



Unseen

Queue: E

Visited: (none)



# BFS Traversal

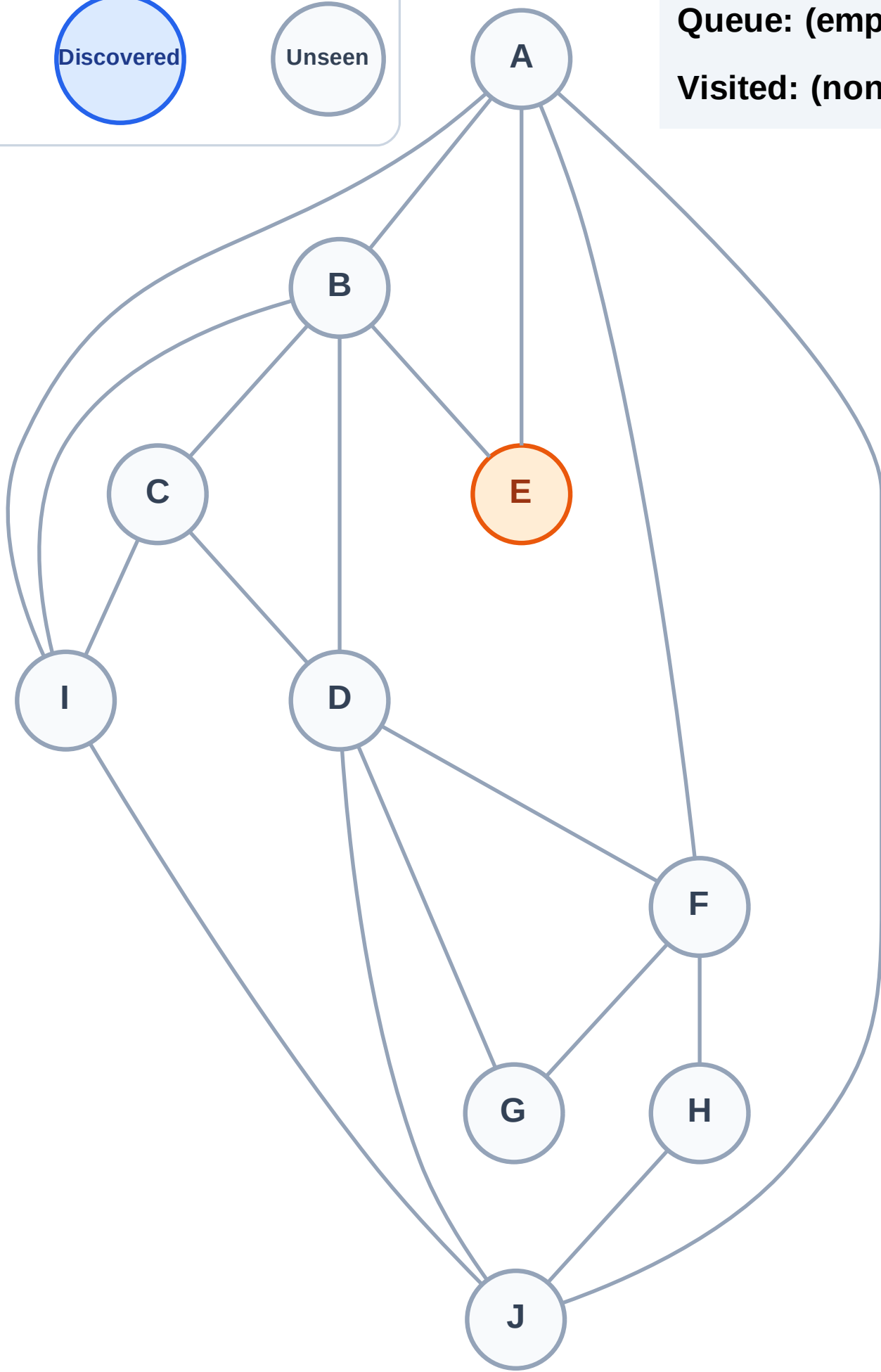
Step 2: Process node E

Legend



Queue: (empty)

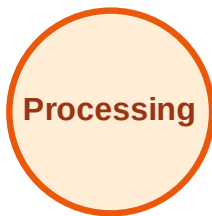
Visited: (none)



# BFS Traversal

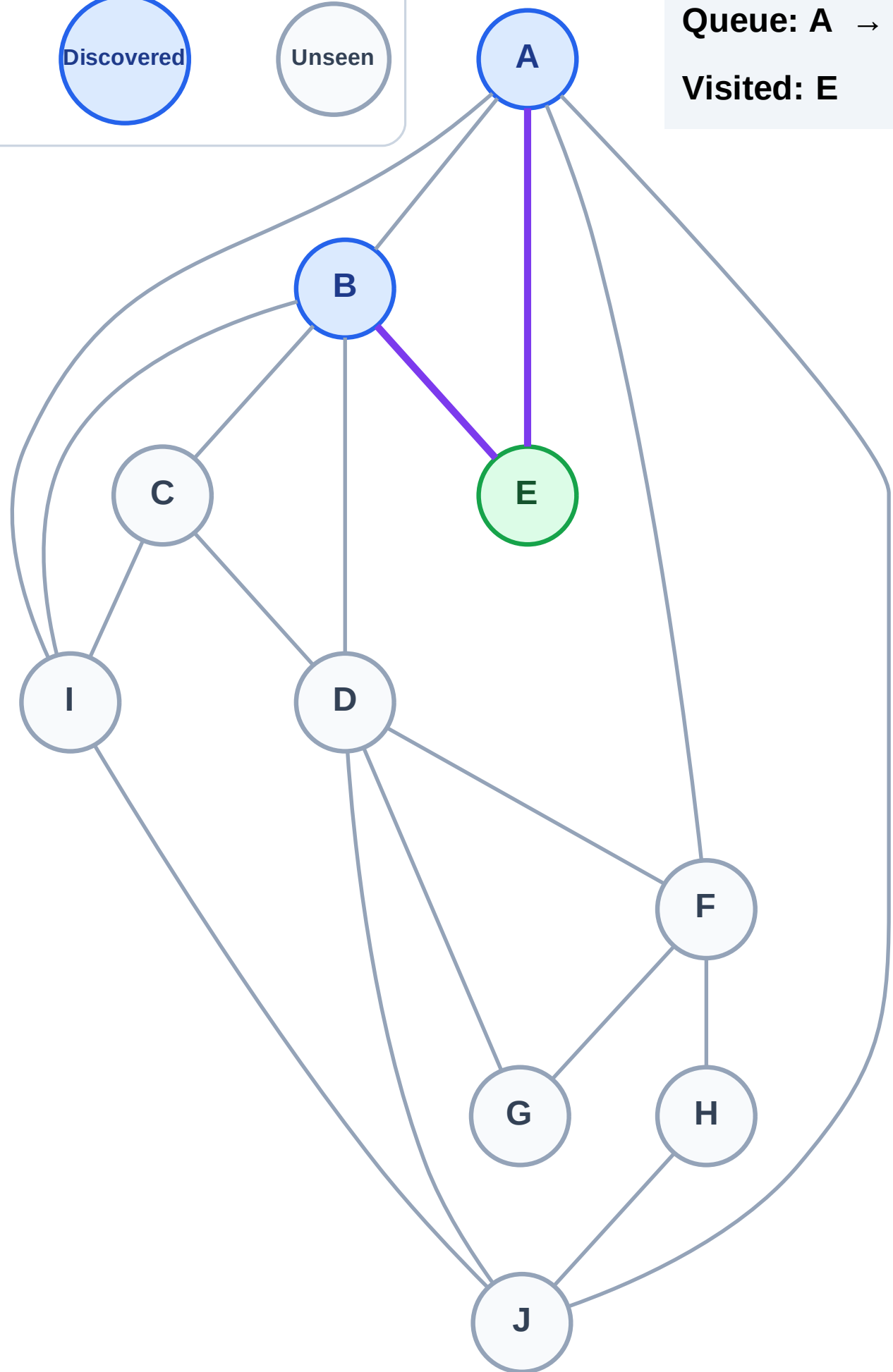
Step 3: Discover A, B from E

## Legend



Queue: A → B

Visited: E



# BFS Traversal

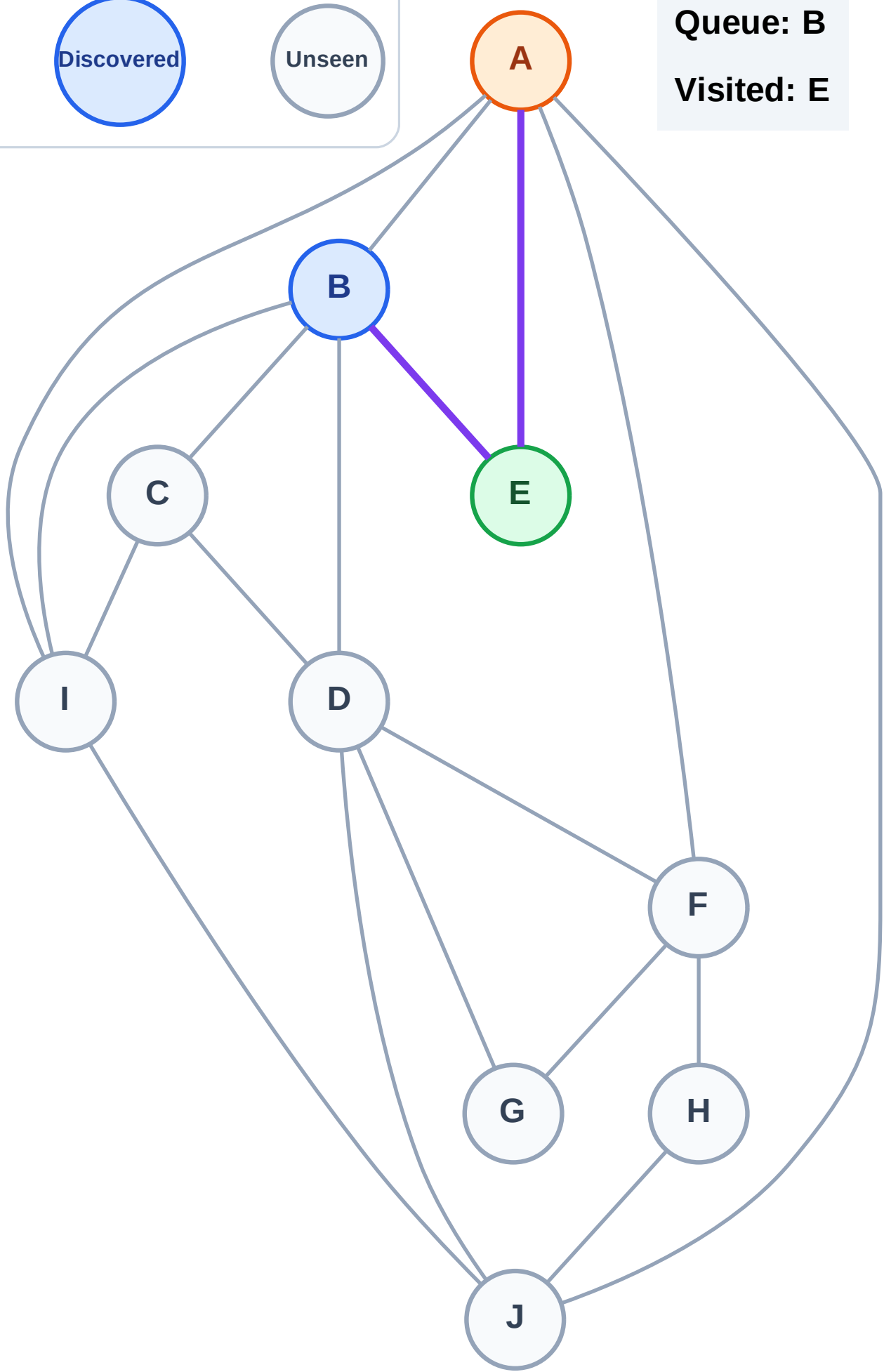
Step 4: Process node A

## Legend



Queue: B

Visited: E



# BFS Traversal

Step 5: Discover F, I, J from A

Legend

Complete

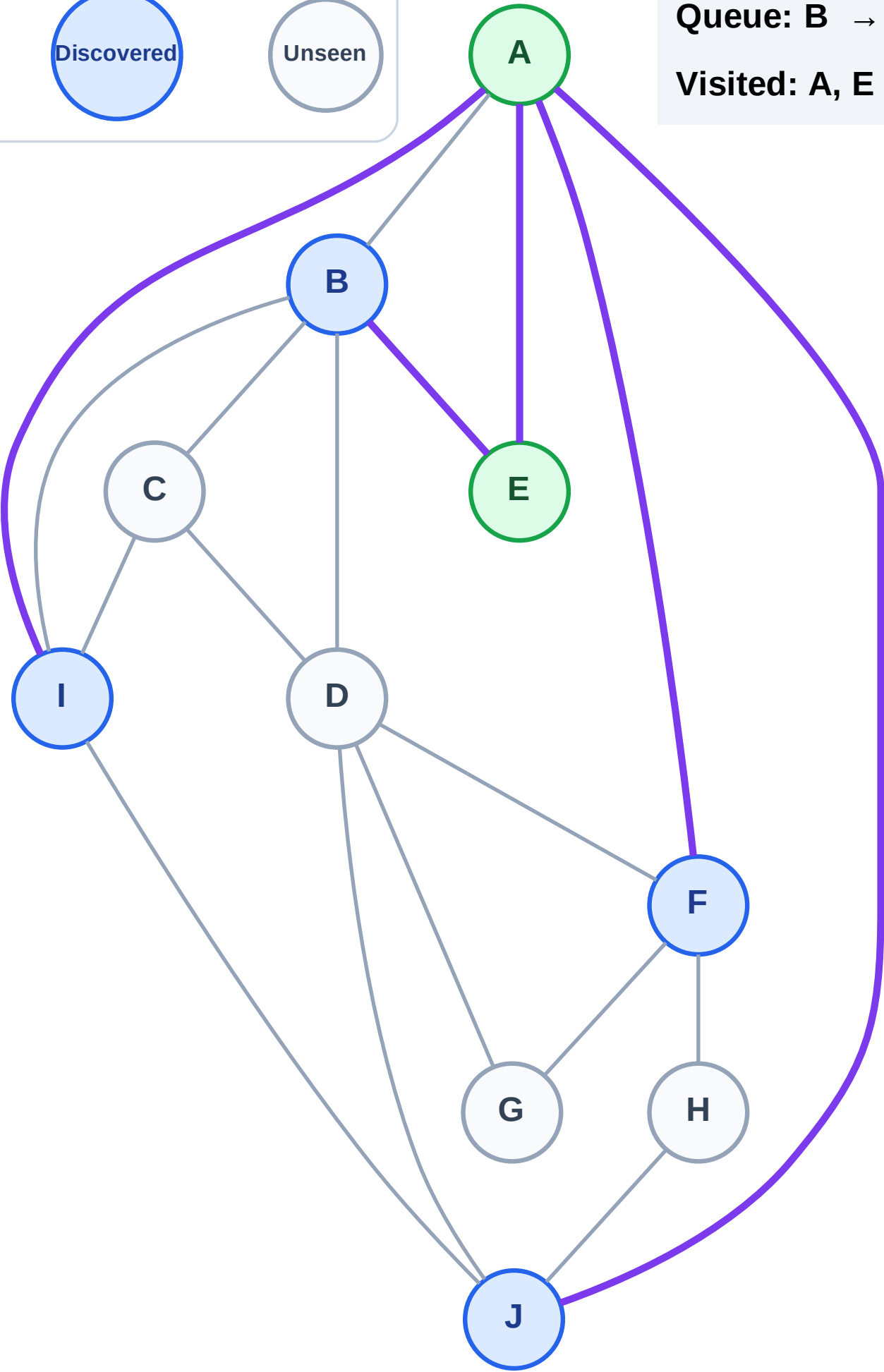
Processing

Discovered

Unseen

Queue: B → F → I → J

Visited: A, E



# BFS Traversal

Step 6: Process node B

Legend

Complete

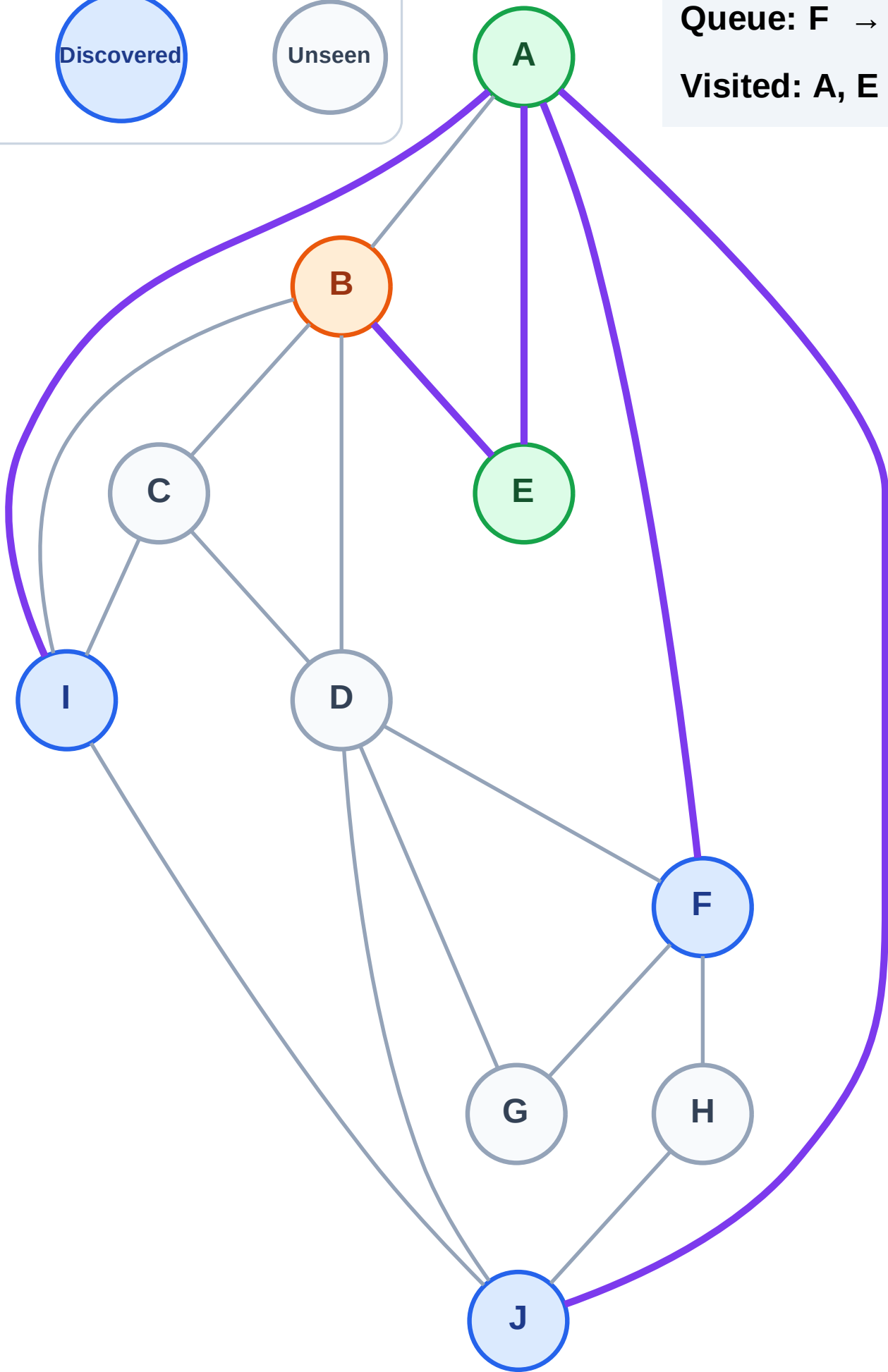
Processing

Discovered

Unseen

Queue: F → I → J

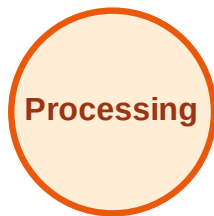
Visited: A, E



# BFS Traversal

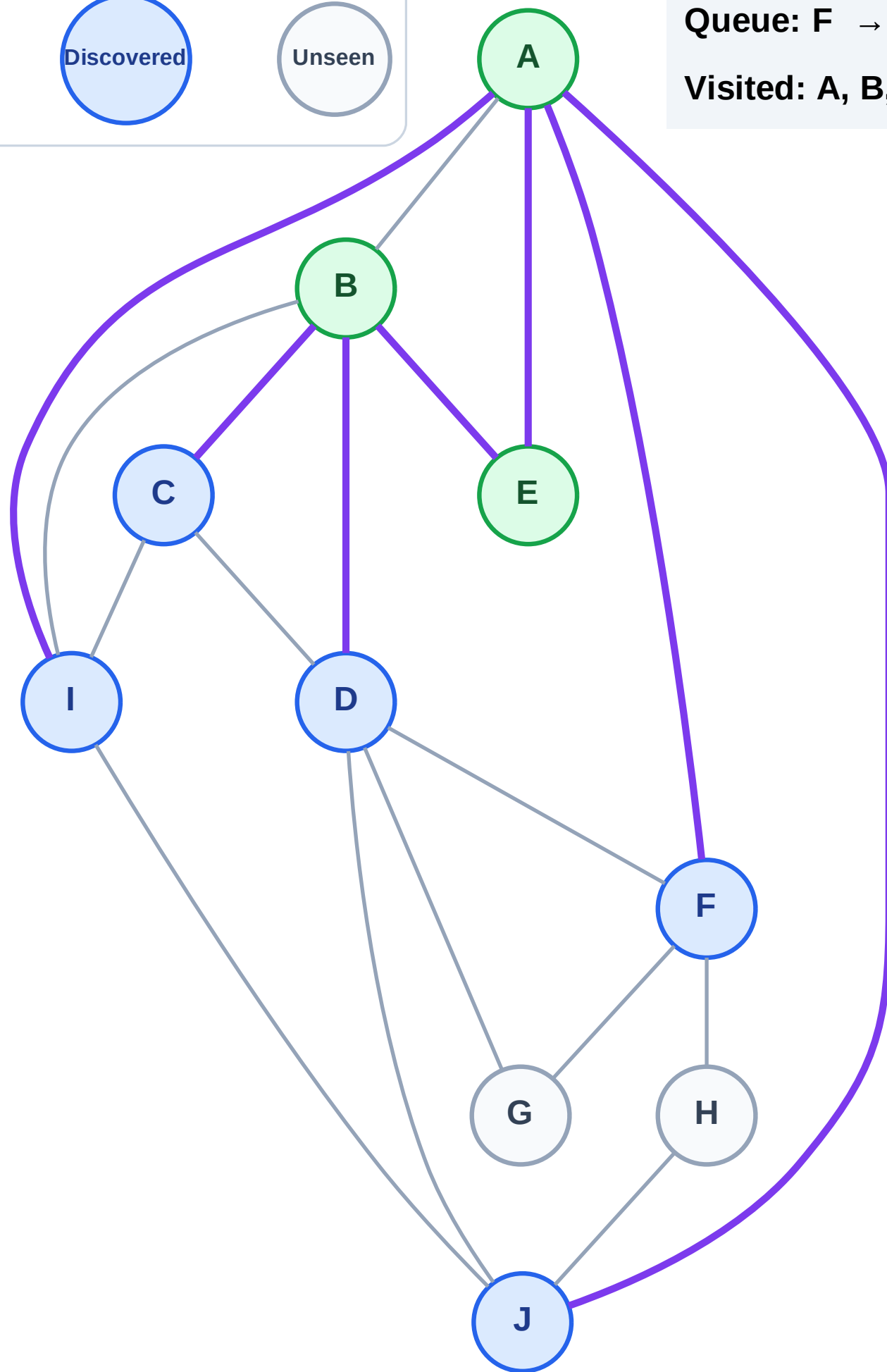
Step 7: Discover C, D from B

## Legend



Queue: F → I → J → C → D

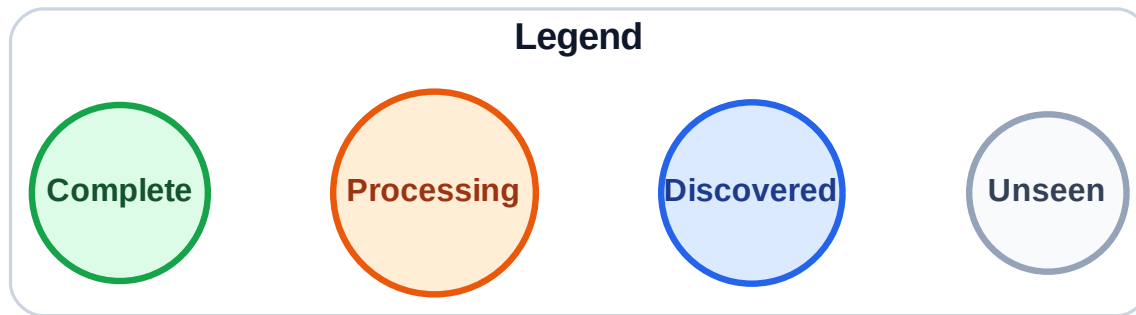
Visited: A, B, E





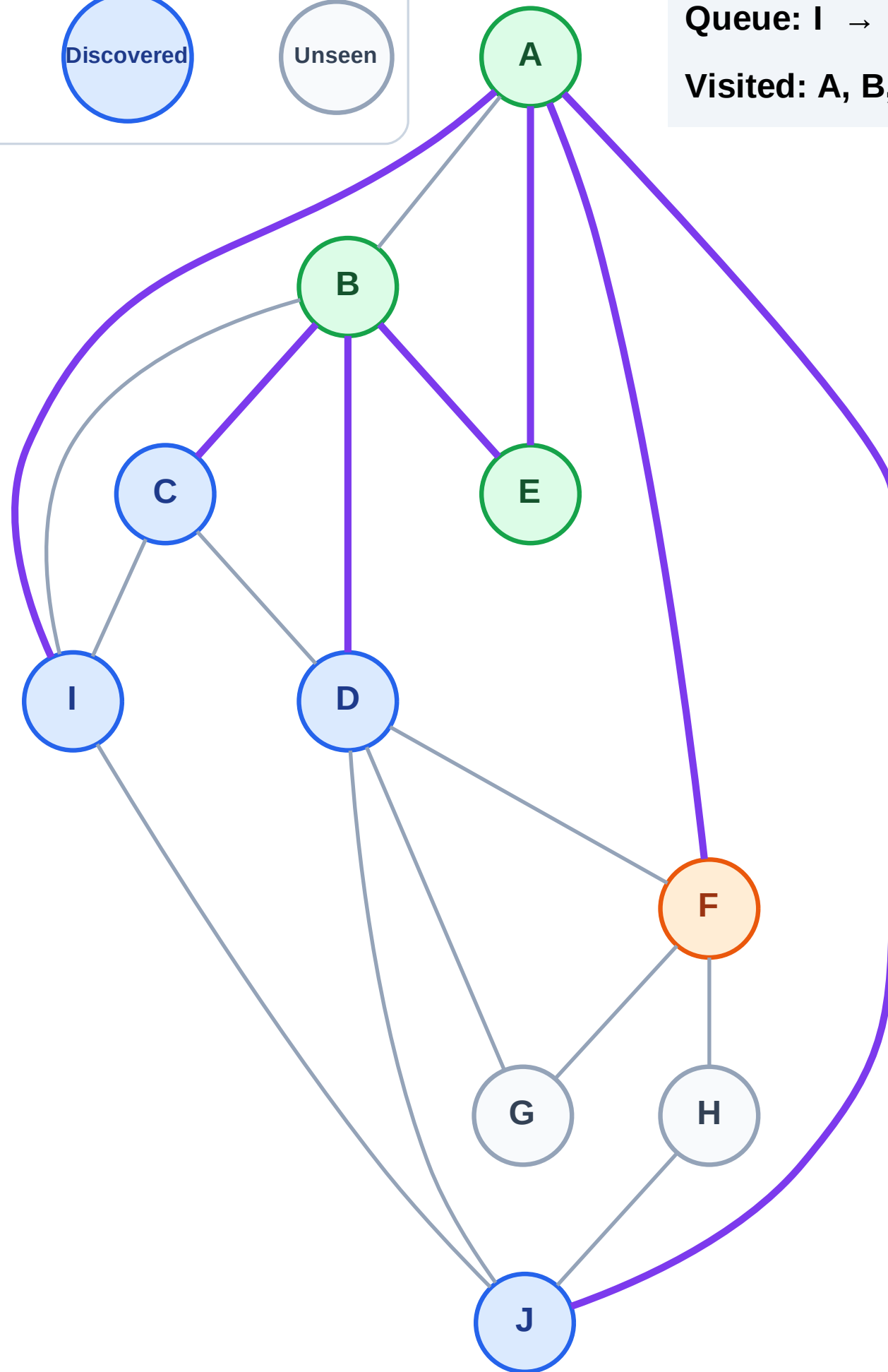
### Step 8: Process node F

### Step 8: Process node F



**Queue: I → J → C → D**

**Visited: A, B, E**





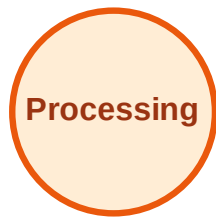
# BFS Traversal

Step 9: Discover G, H from F

## Legend



Complete



Processing



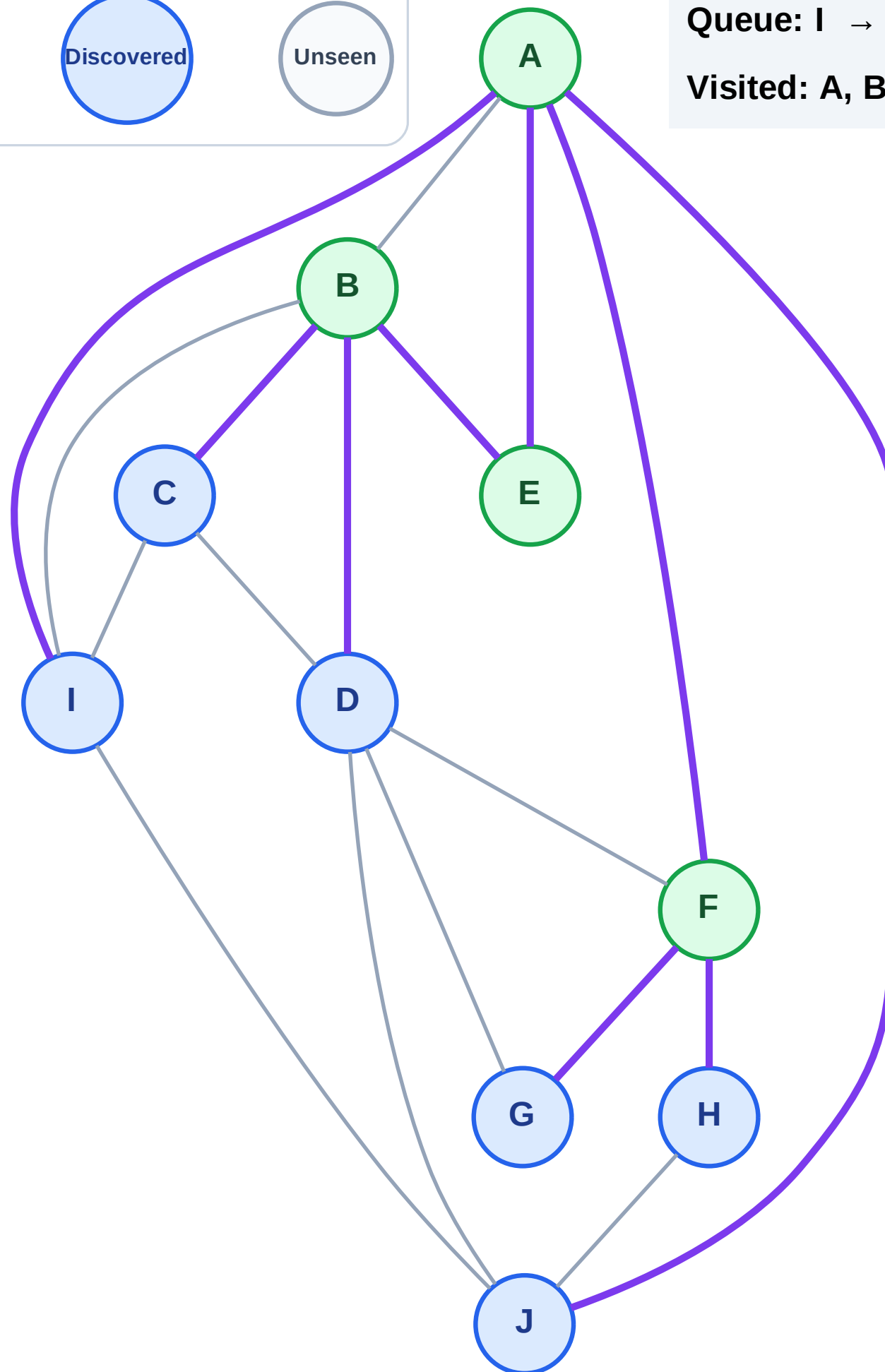
Discovered



Unseen

Queue: I → J → C → D → G → H

Visited: A, B, E, F



# BFS Traversal

Step 10: Process node I

Legend

Complete

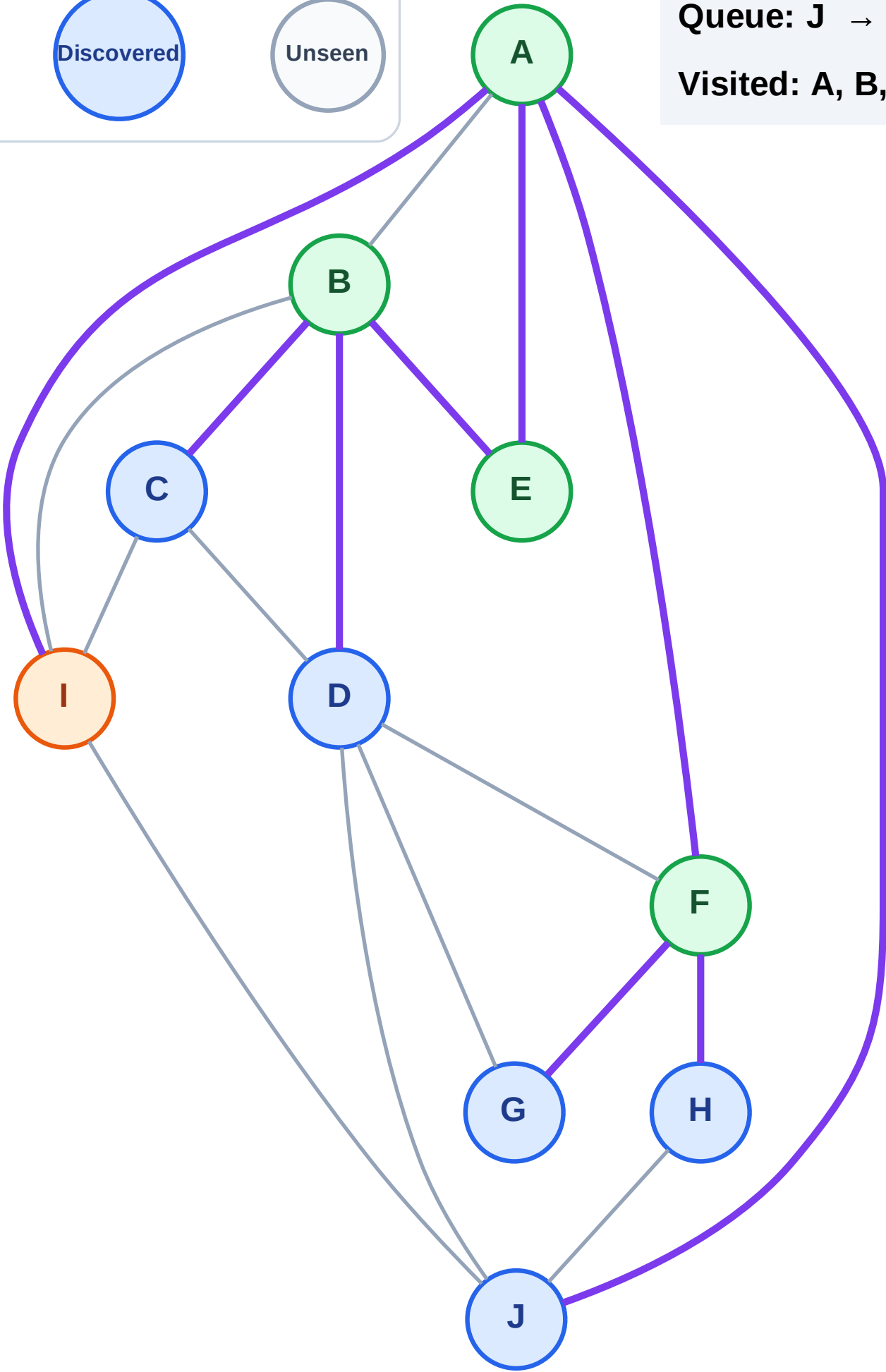
Processing

Discovered

Unseen

Queue: J → C → D → G → H

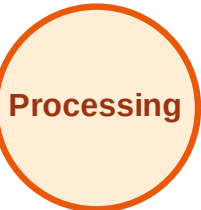
Visited: A, B, E, F



# BFS Traversal

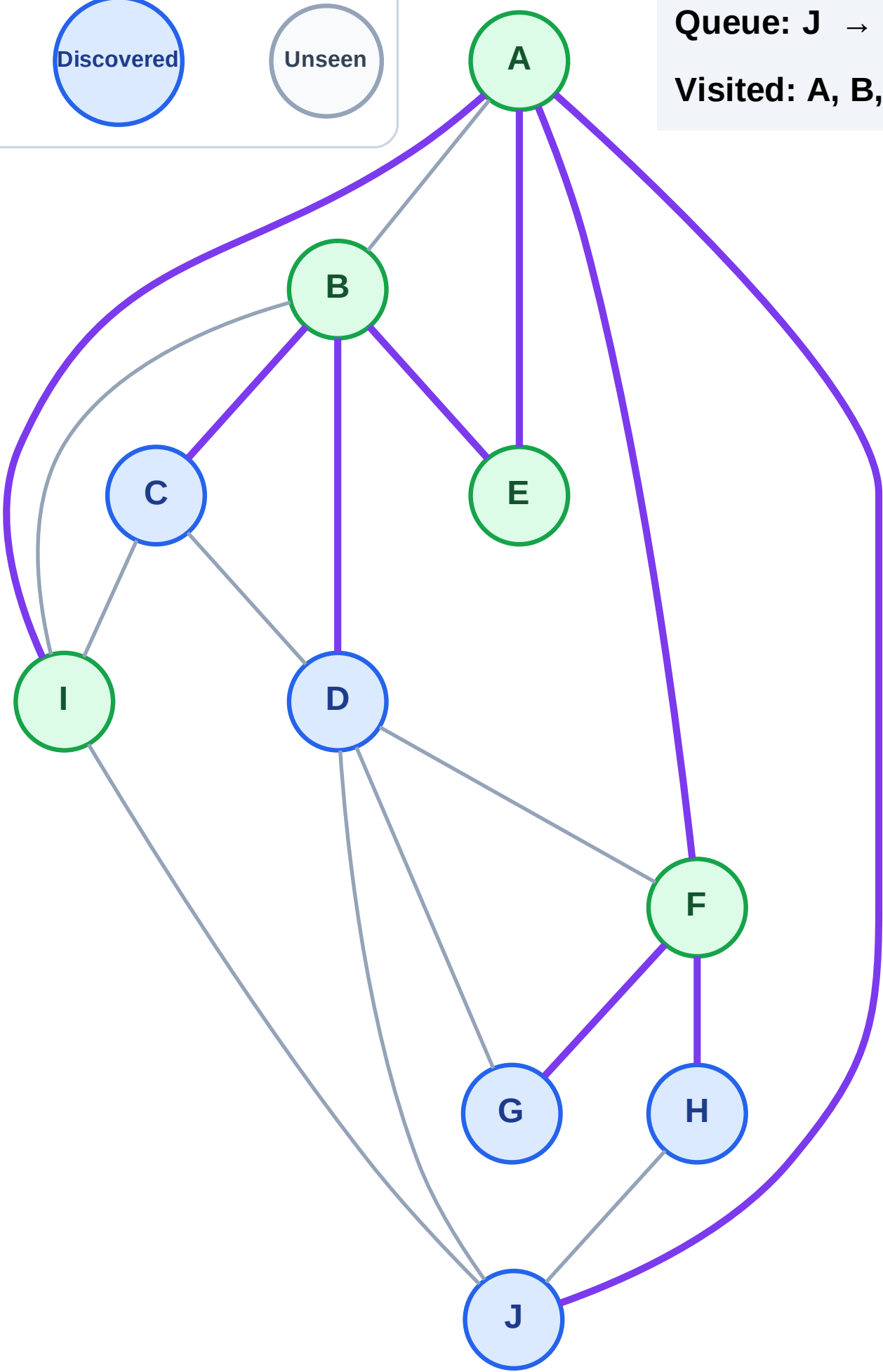
Step 11: No new neighbors from I

## Legend



Queue: J → C → D → G → H

Visited: A, B, E, F, I



# BFS Traversal

Step 12: Process node J

Legend

Complete

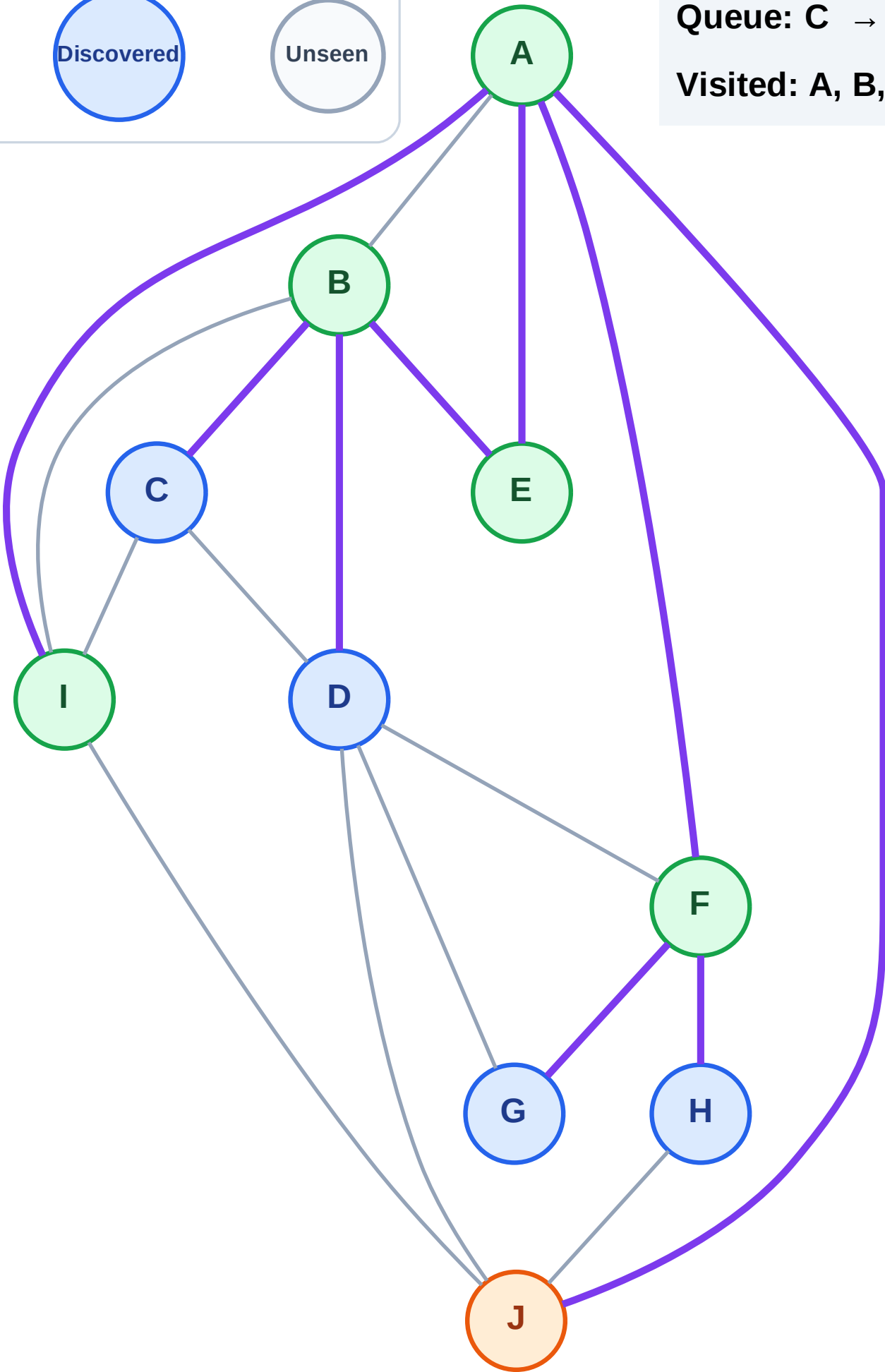
Processing

Discovered

Unseen

Queue: C → D → G → H

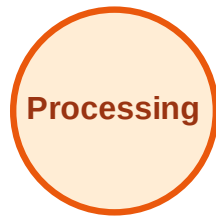
Visited: A, B, E, F, I



# BFS Traversal

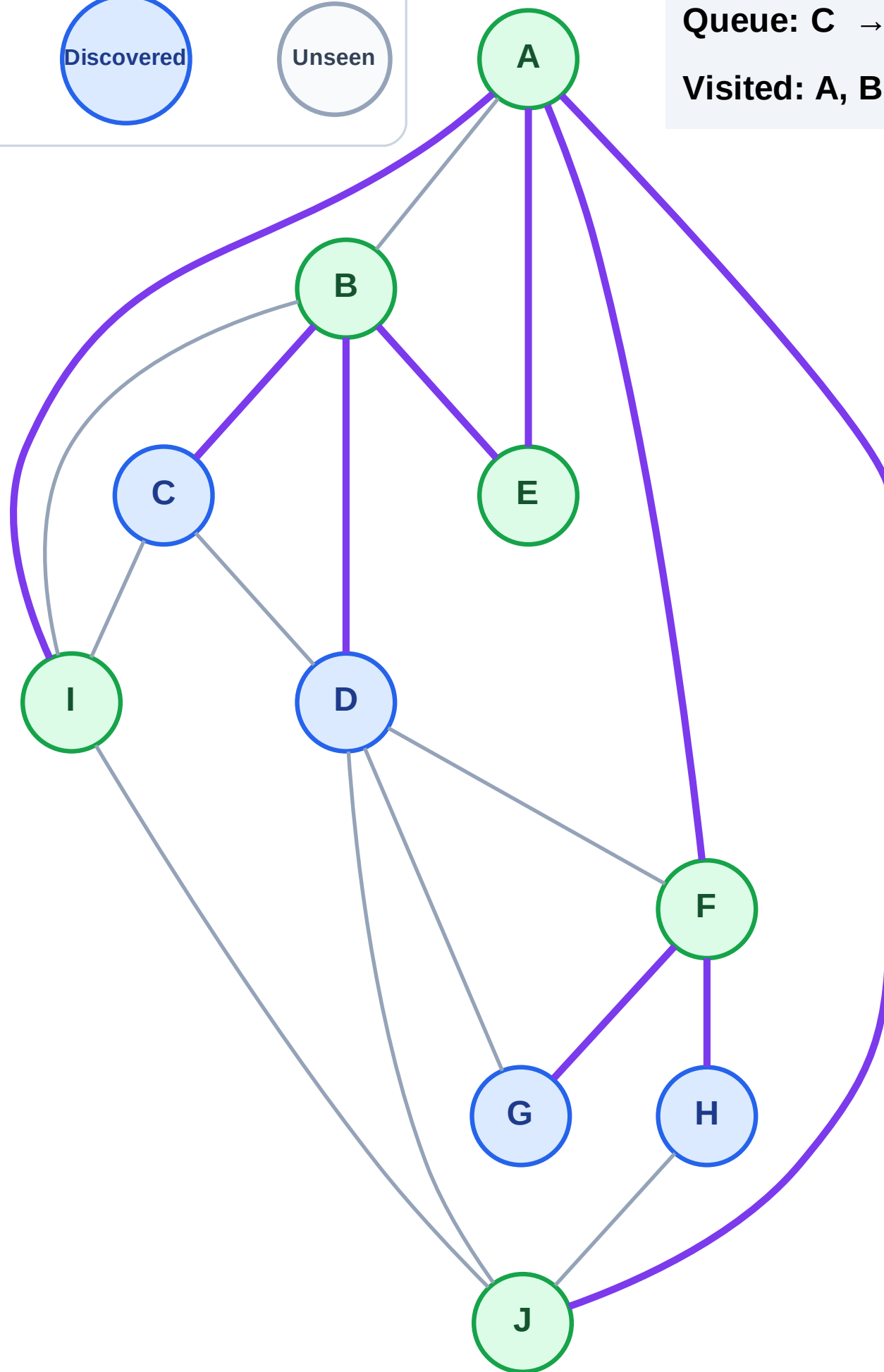
Step 13: No new neighbors from J

## Legend



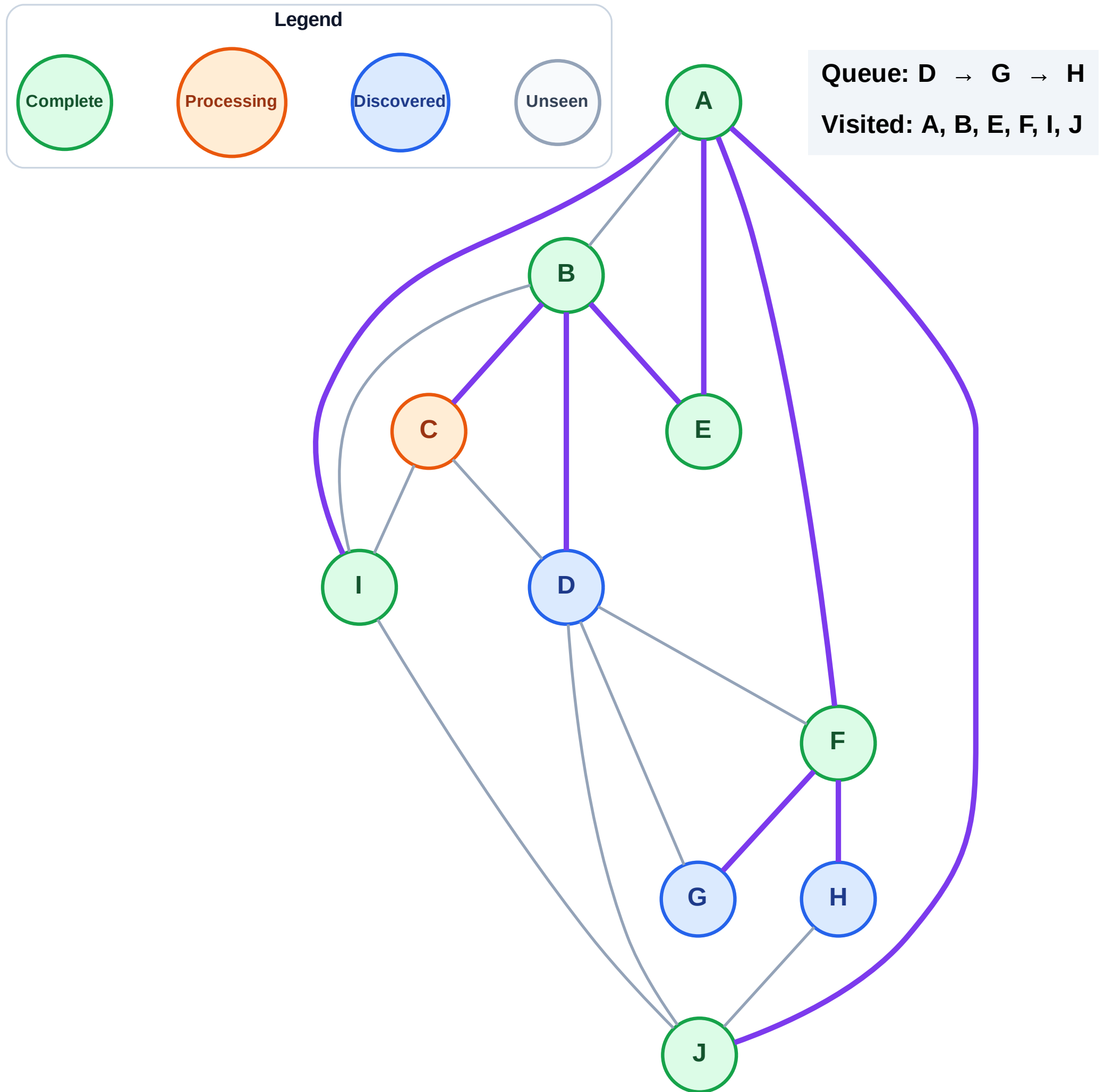
Queue: C → D → G → H

Visited: A, B, E, F, I, J



### Step 14: Process node C

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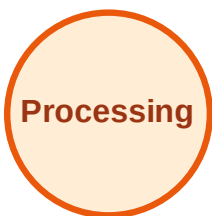




# BFS Traversal

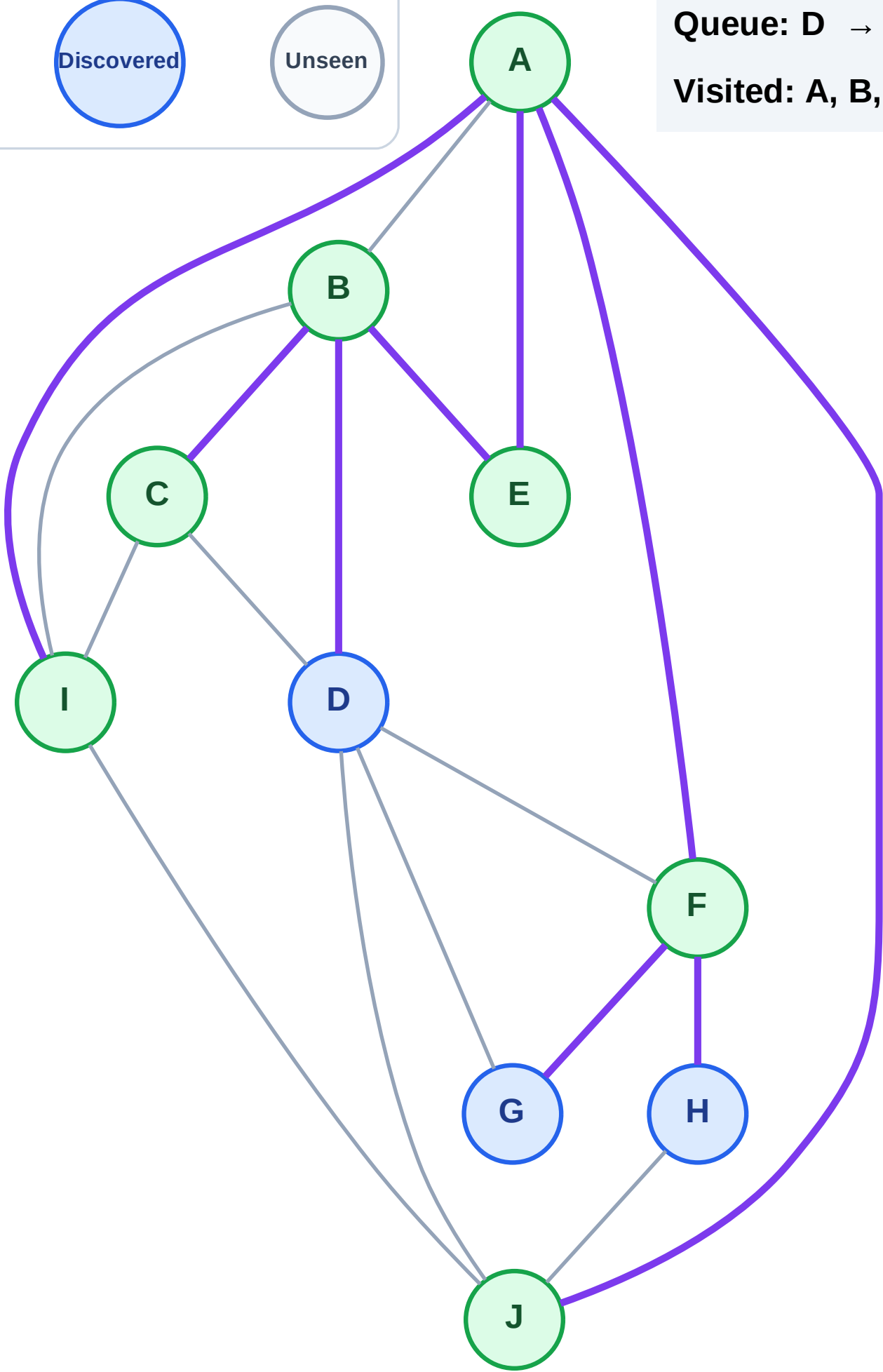
Step 15: No new neighbors from C

## Legend



Queue: D → G → H

Visited: A, B, C, E, F, I, J



# BFS Traversal

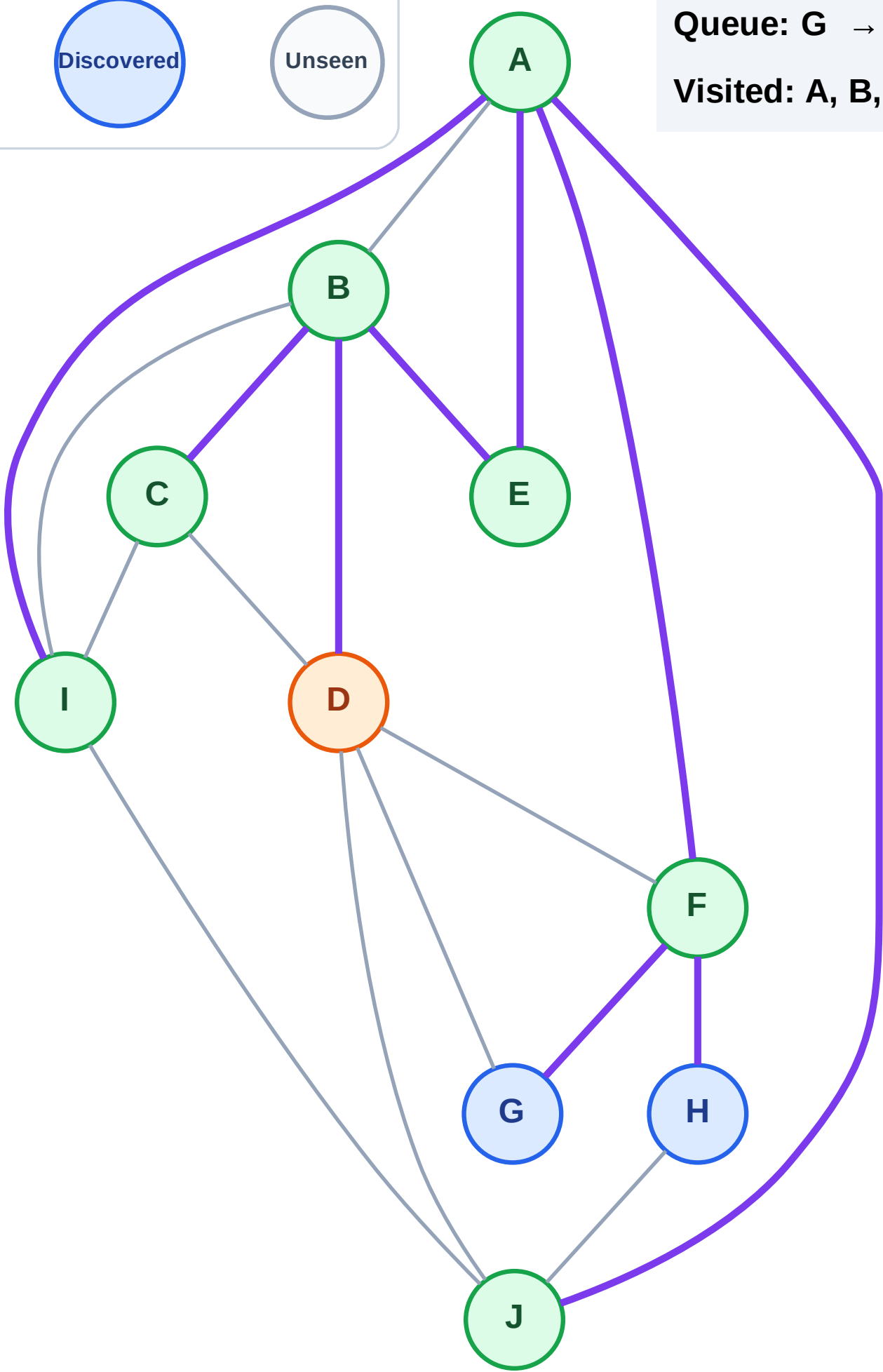
Step 16: Process node D

## Legend



Queue: G → H

Visited: A, B, C, E, F, I, J





# BFS Traversal

Step 17: No new neighbors from D

Legend

Complete

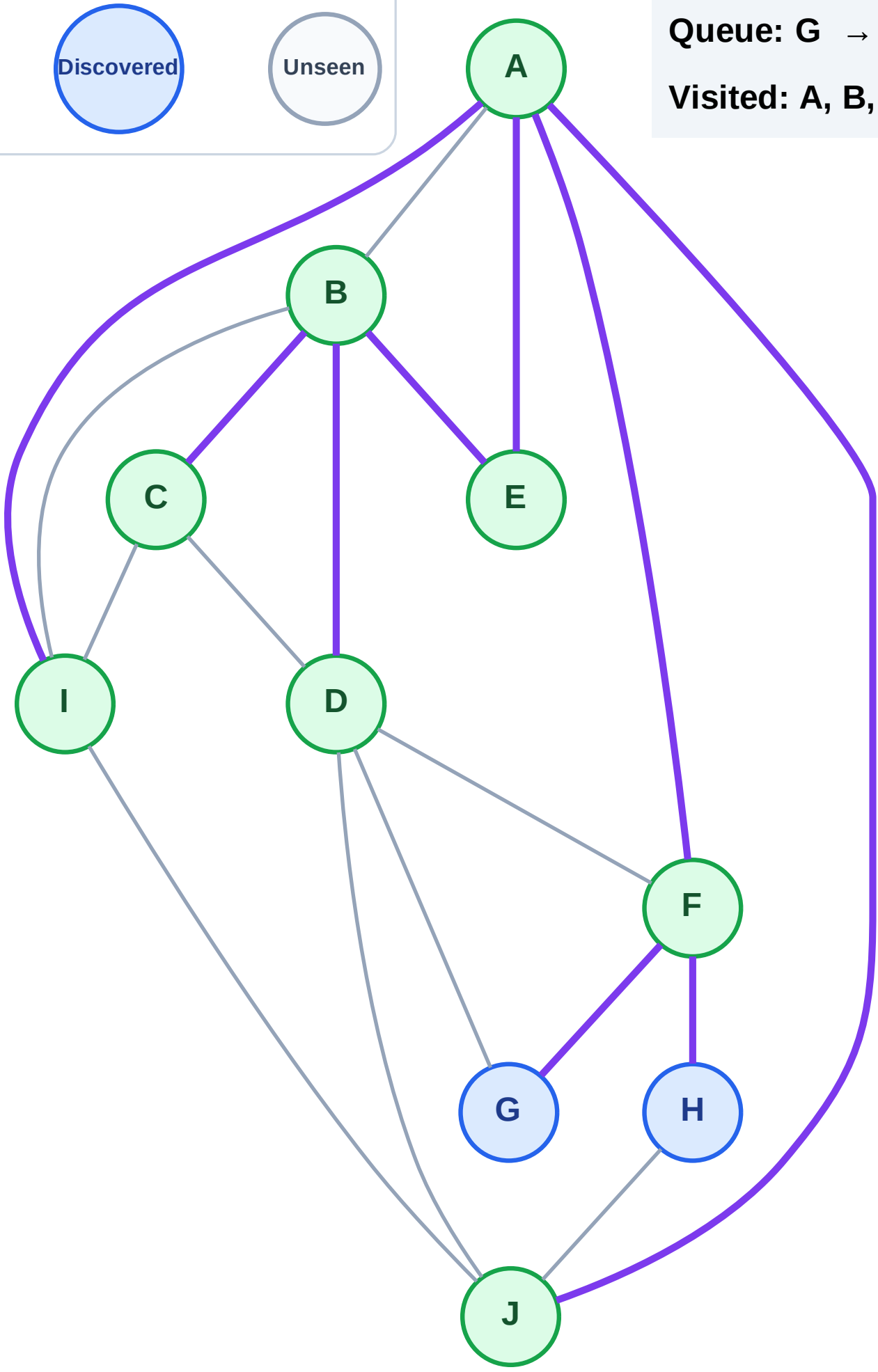
Processing

Discovered

Unseen

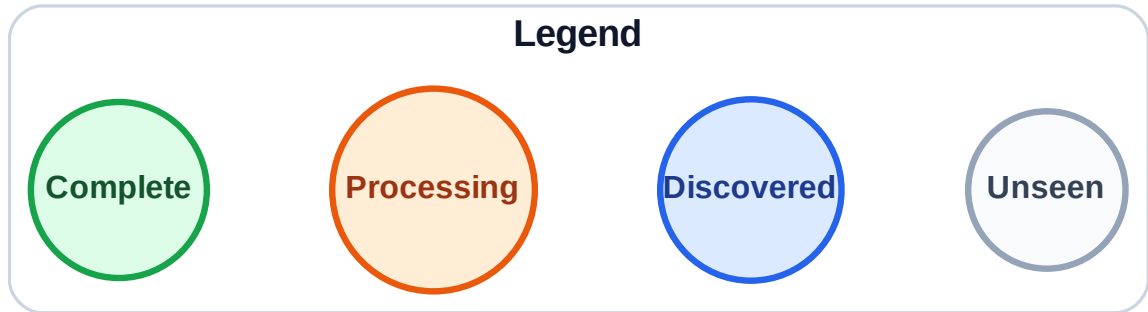
Queue: G → H

Visited: A, B, C, D, E, F, I, J



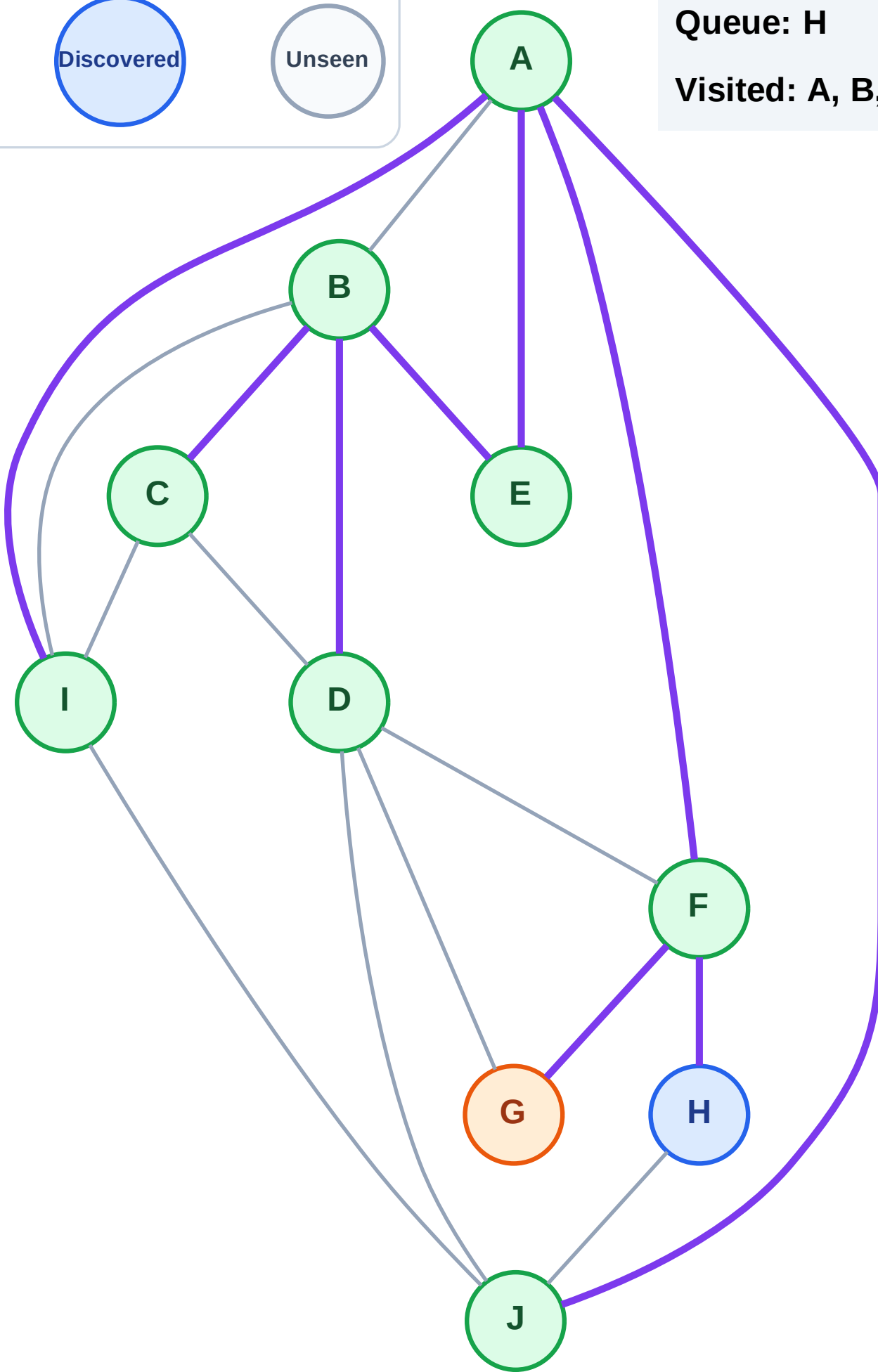
## Step 18: Process node G

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**Queue: H**

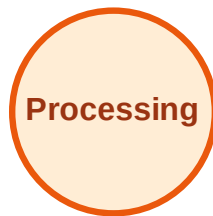
**Visited: A, B, C, D, E, F, I, J**



# BFS Traversal

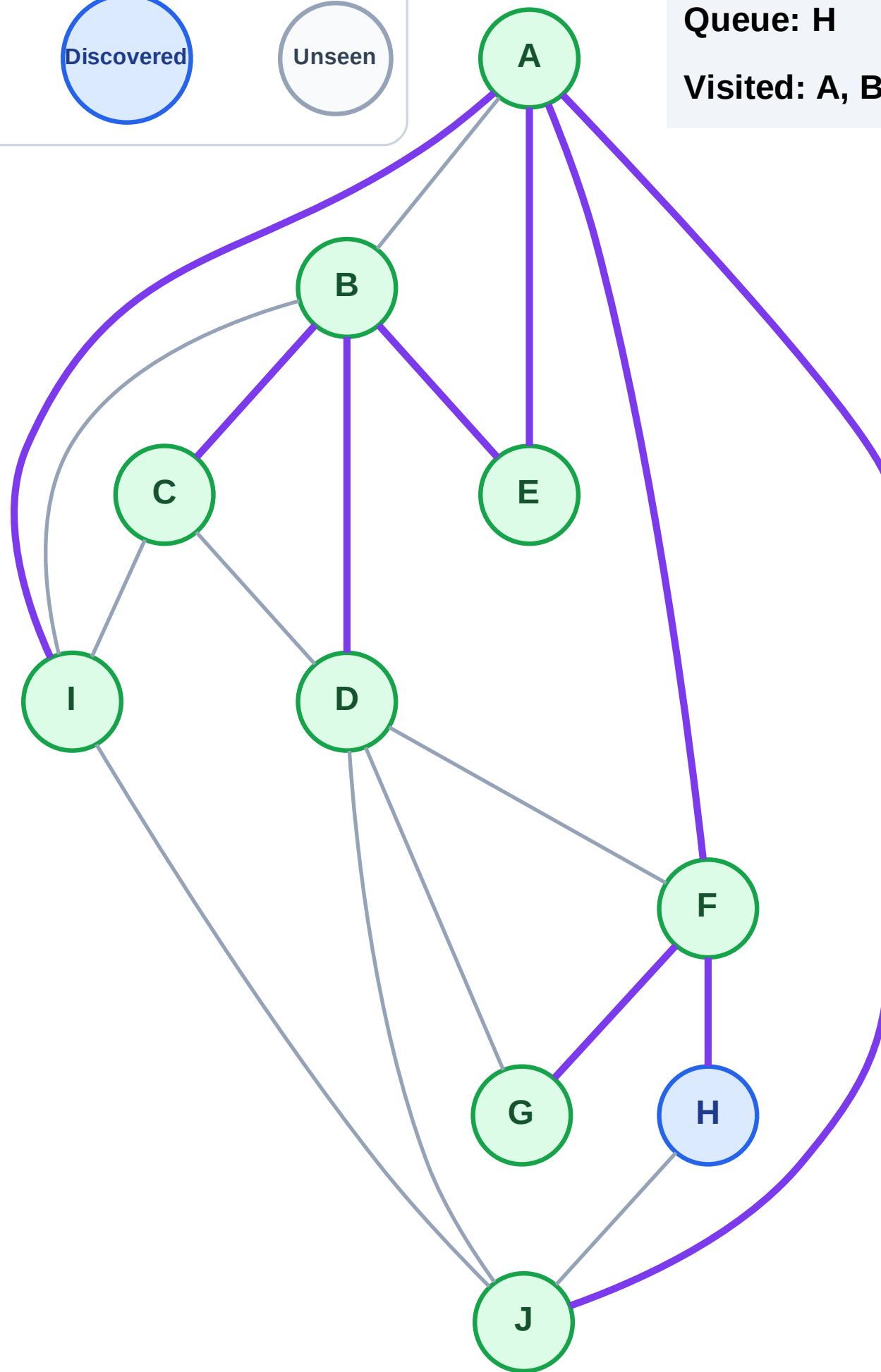
Step 19: No new neighbors from G

## Legend



Queue: H

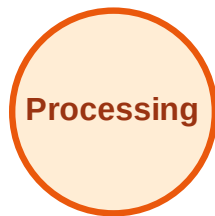
Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

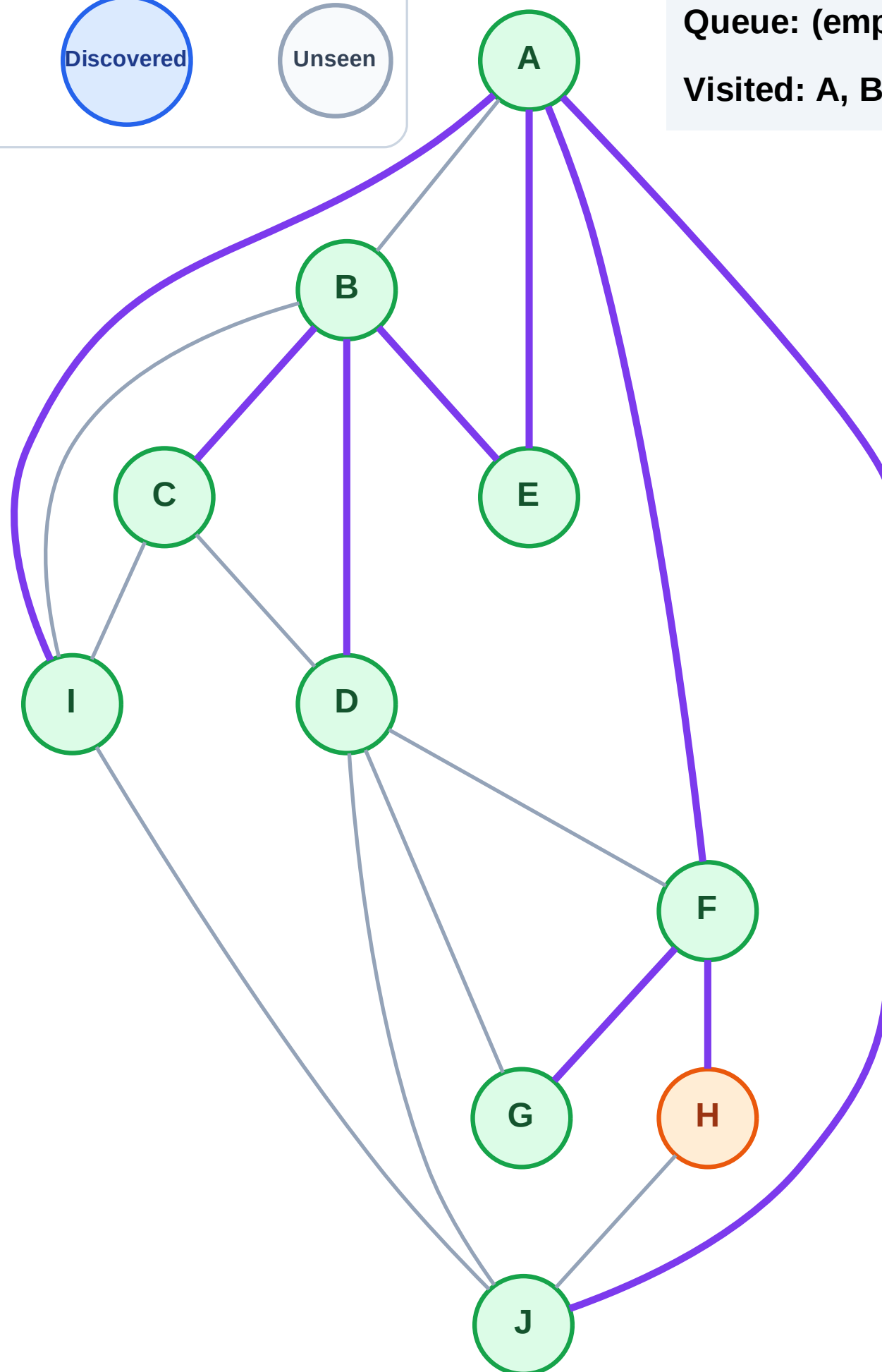
Step 20: Process node H

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, I, J



# BFS Traversal

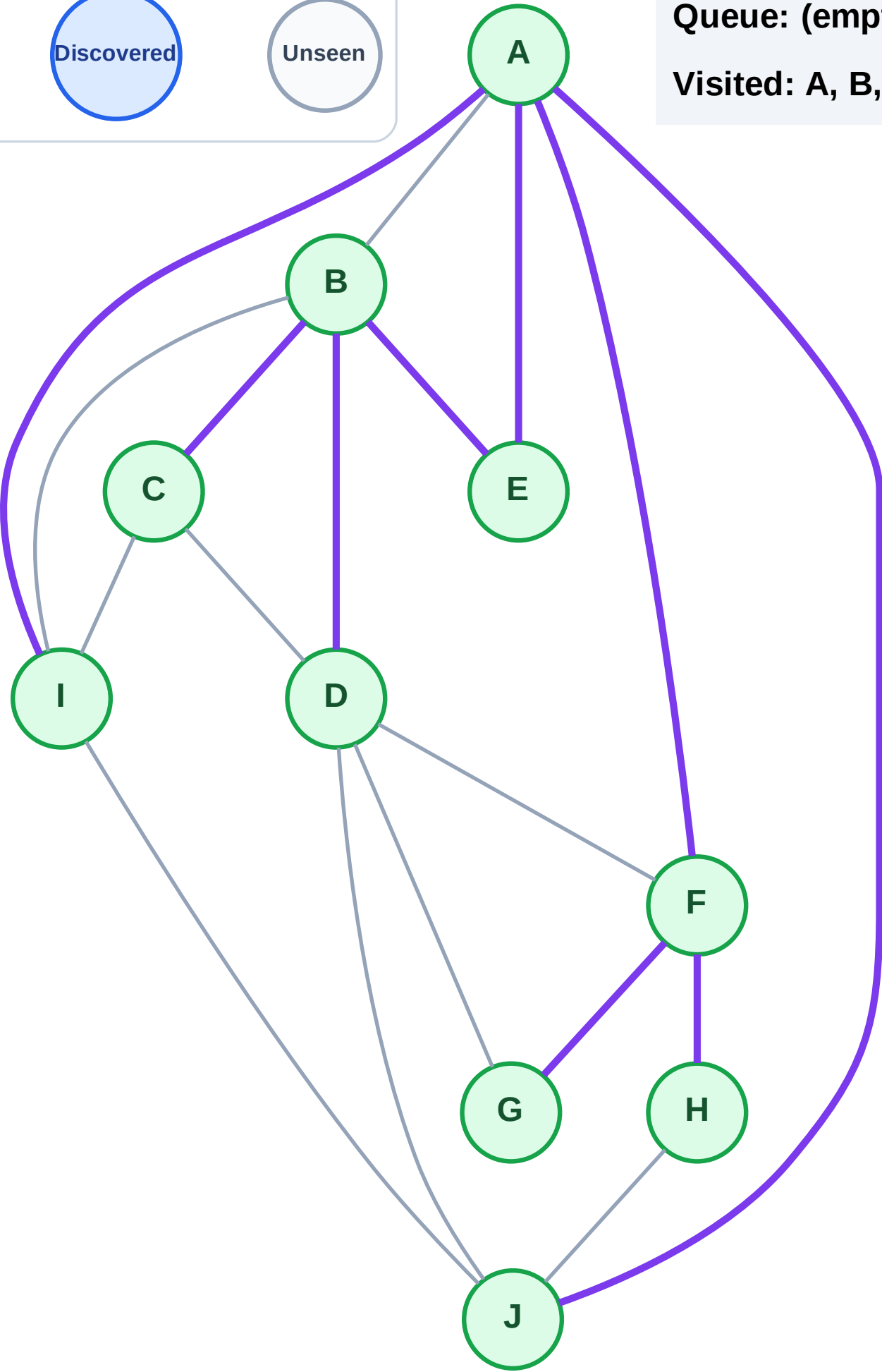
Step 21: No new neighbors from H

## Legend



Queue: (empty)

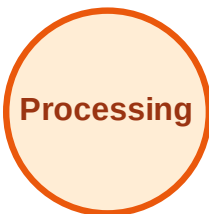
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

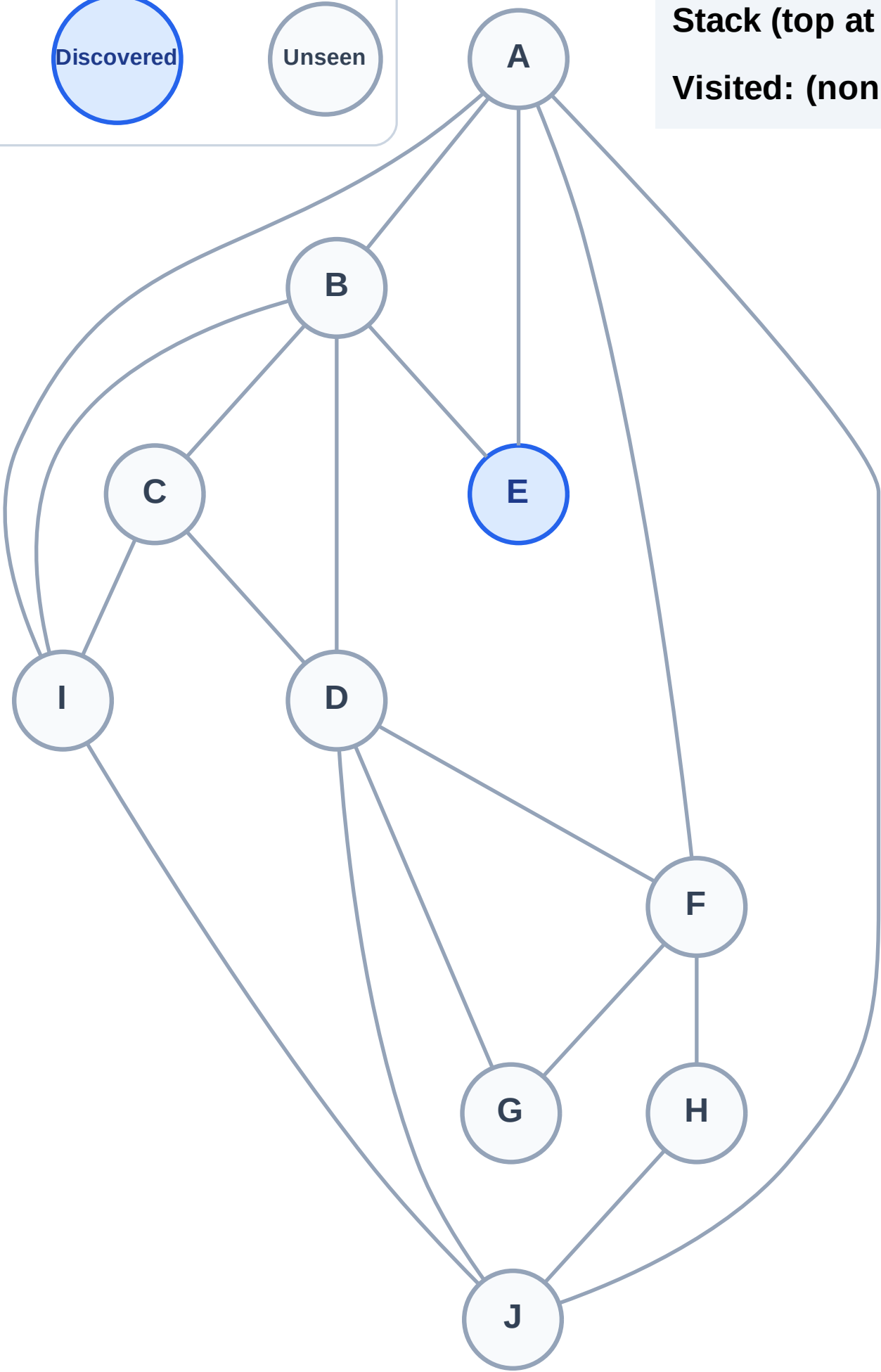
Step 1: Start at E

## Legend



Stack (top at right): E

Visited: (none)





# DFS Traversal

Step 2: Process node E

Legend

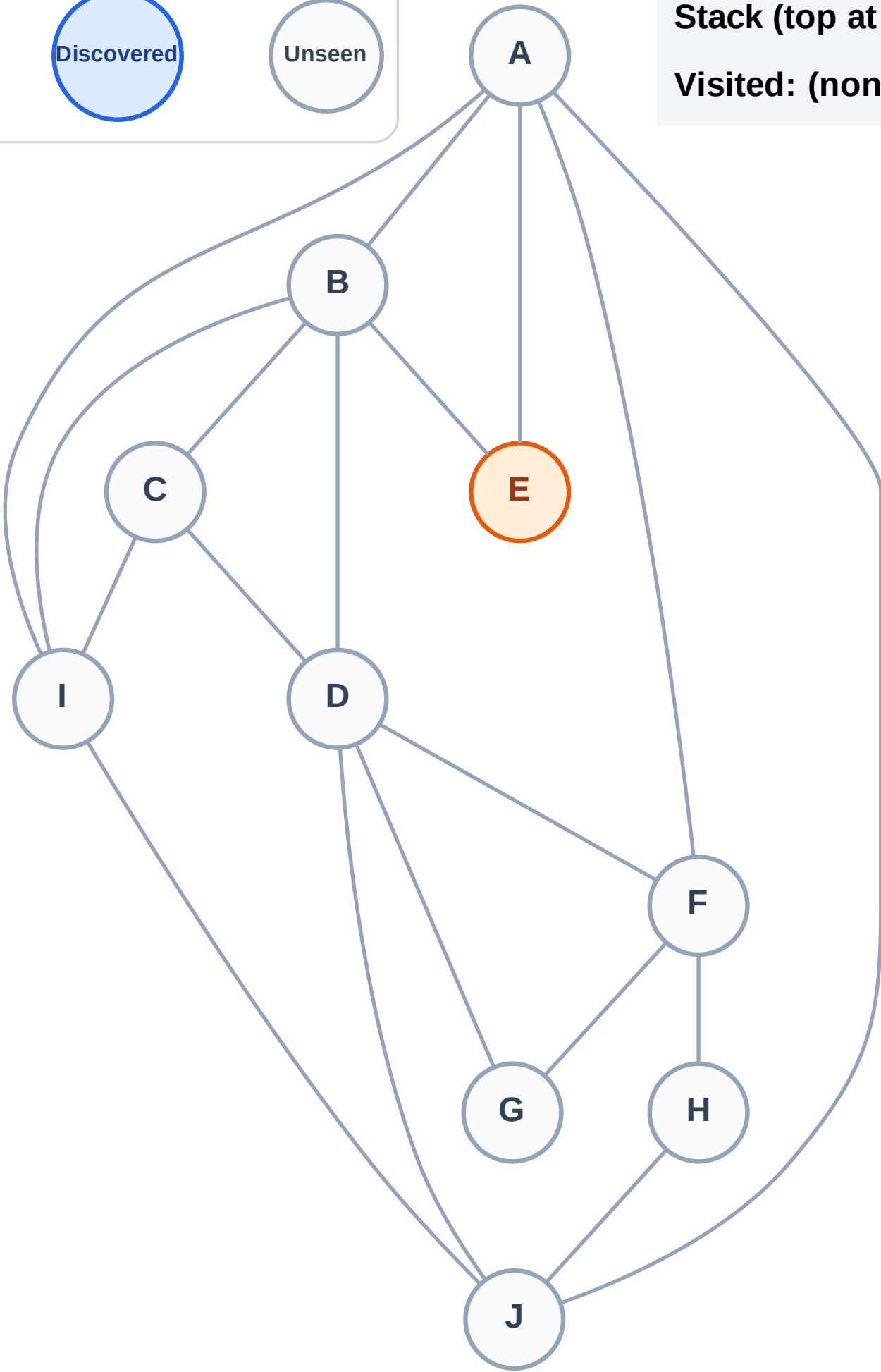
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

Step 3: Discover B, A from E

Legend

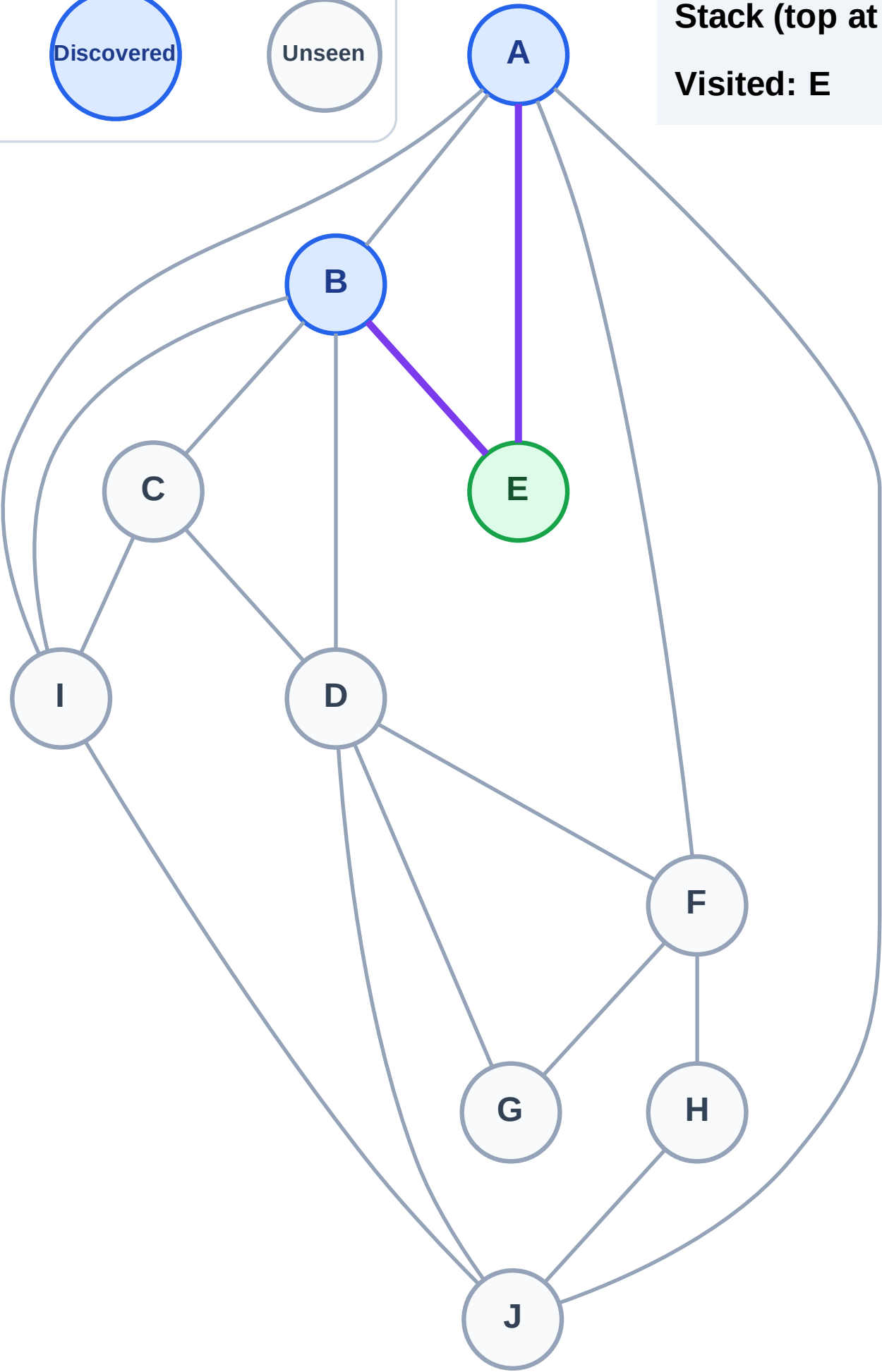
Complete

Processing

Discovered

Unseen

Stack (top at right): B | A  
Visited: E





# DFS Traversal

Step 4: Process node A

Legend

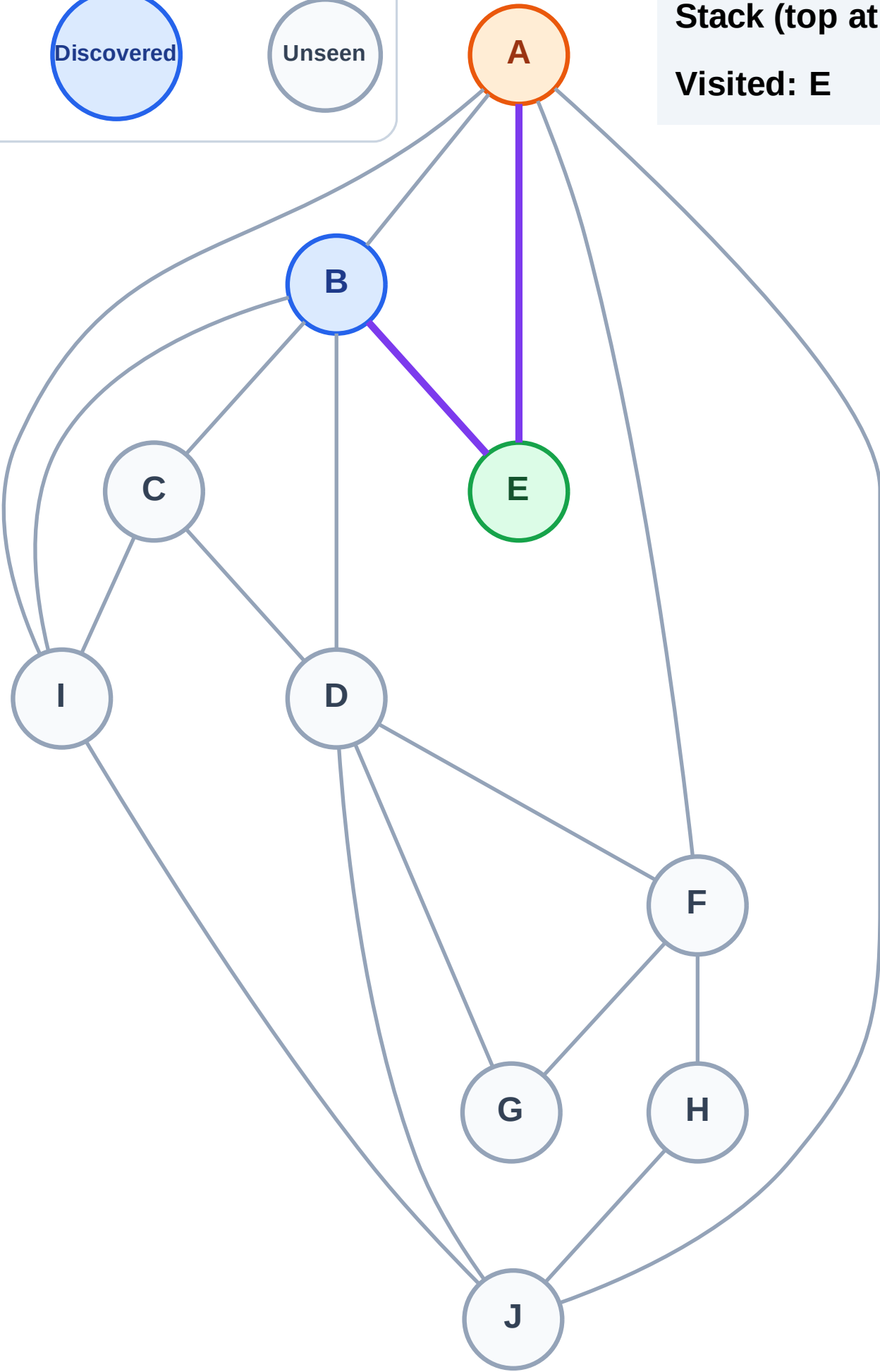
Complete

Processing

Discovered

Unseen

Stack (top at right): B  
Visited: E



# DFS Traversal

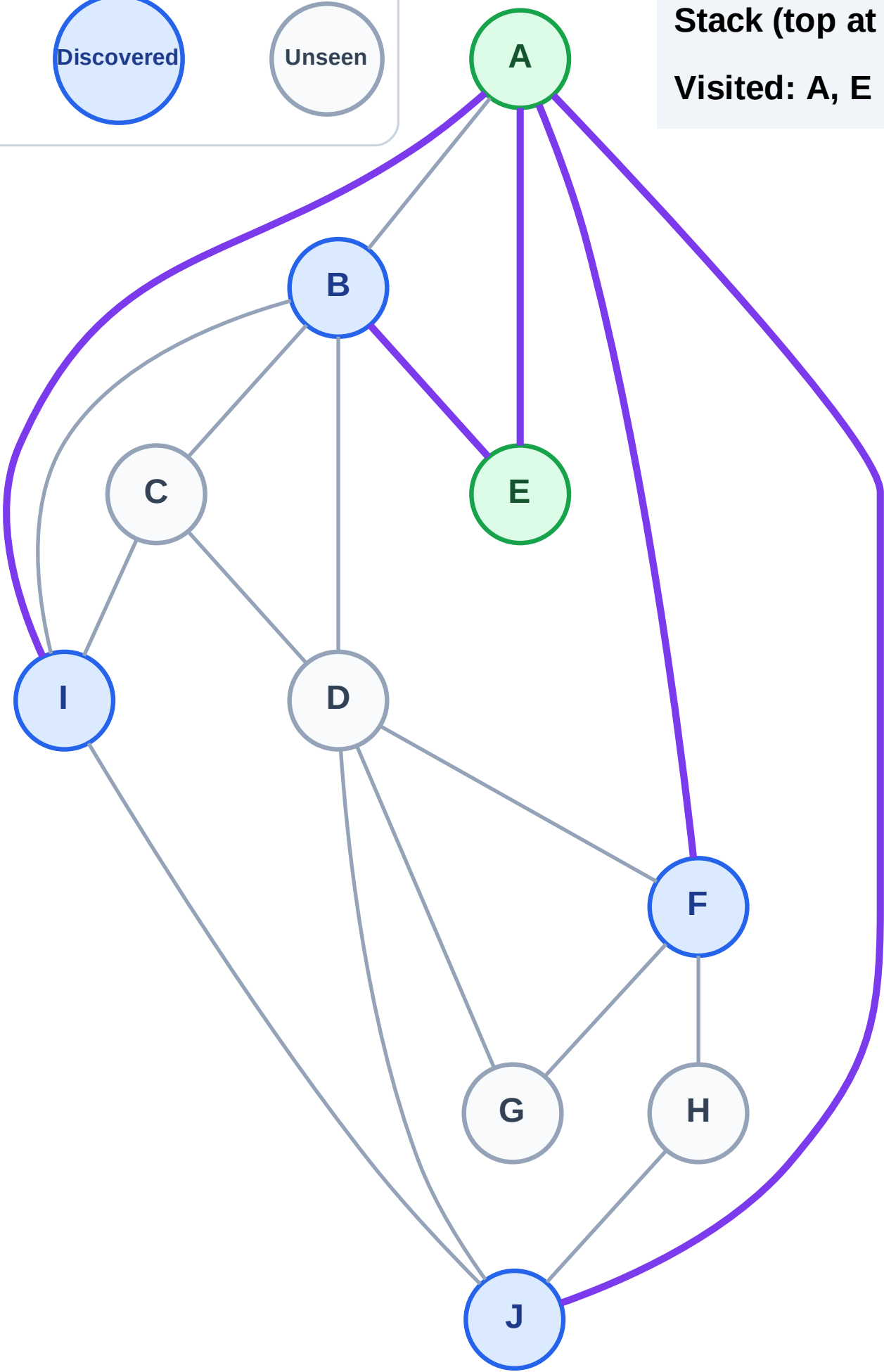
Step 5: Discover J, I, F from A

## Legend



Stack (top at right): B | J | I | F

Visited: A, E



# DFS Traversal

Step 6: Process node F

## Legend

Complete

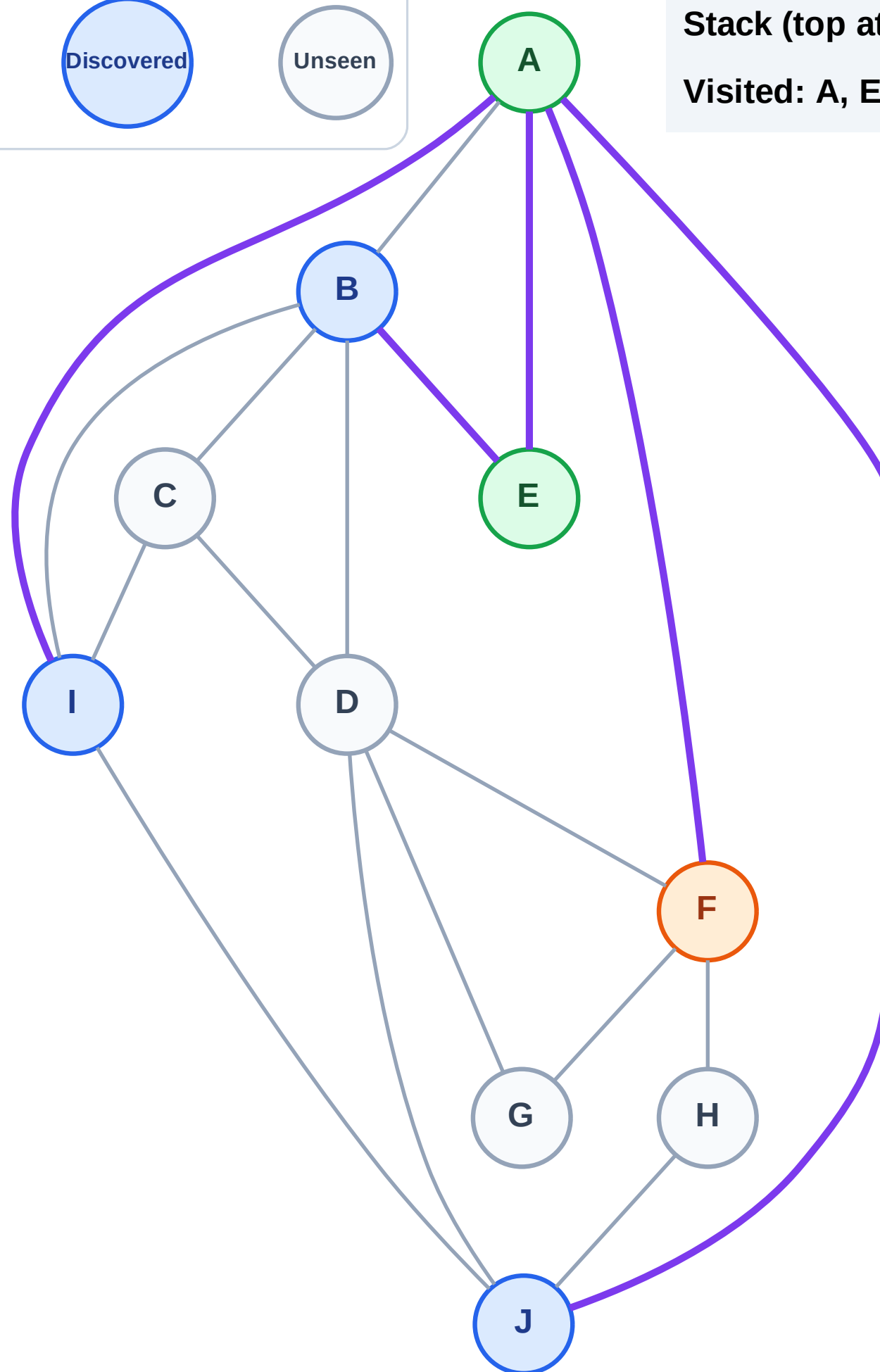
Processing

Discovered

Unseen

Stack (top at right): B | J | I

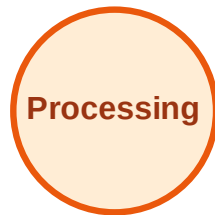
Visited: A, E



# DFS Traversal

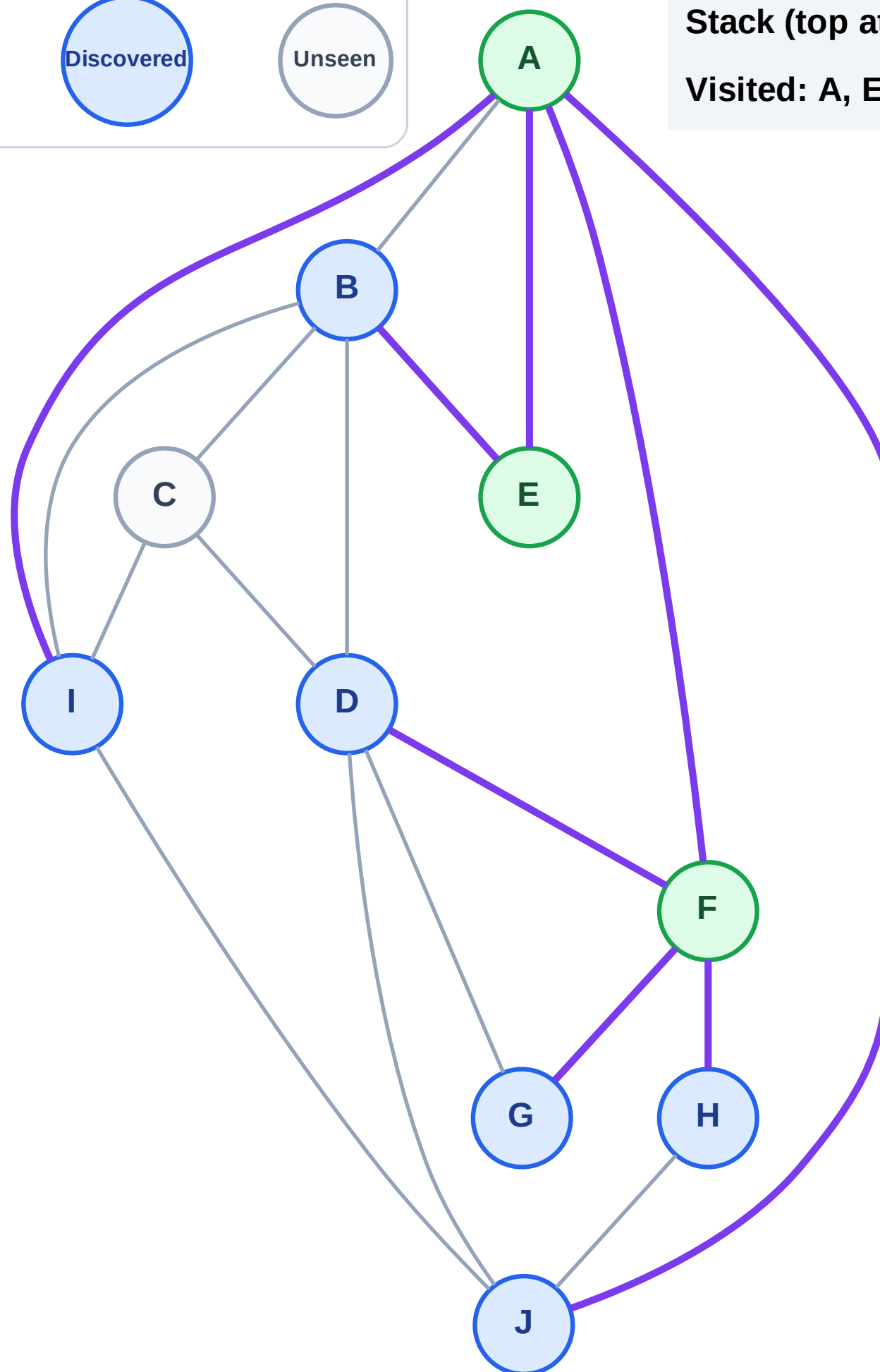
Step 7: Discover H, G, D from F

## Legend



Stack (top at right): B | J | I | H | G | D

Visited: A, E, F



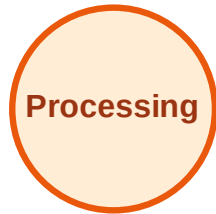
# DFS Traversal

Step 8: Process node D

## Legend



Complete



Processing



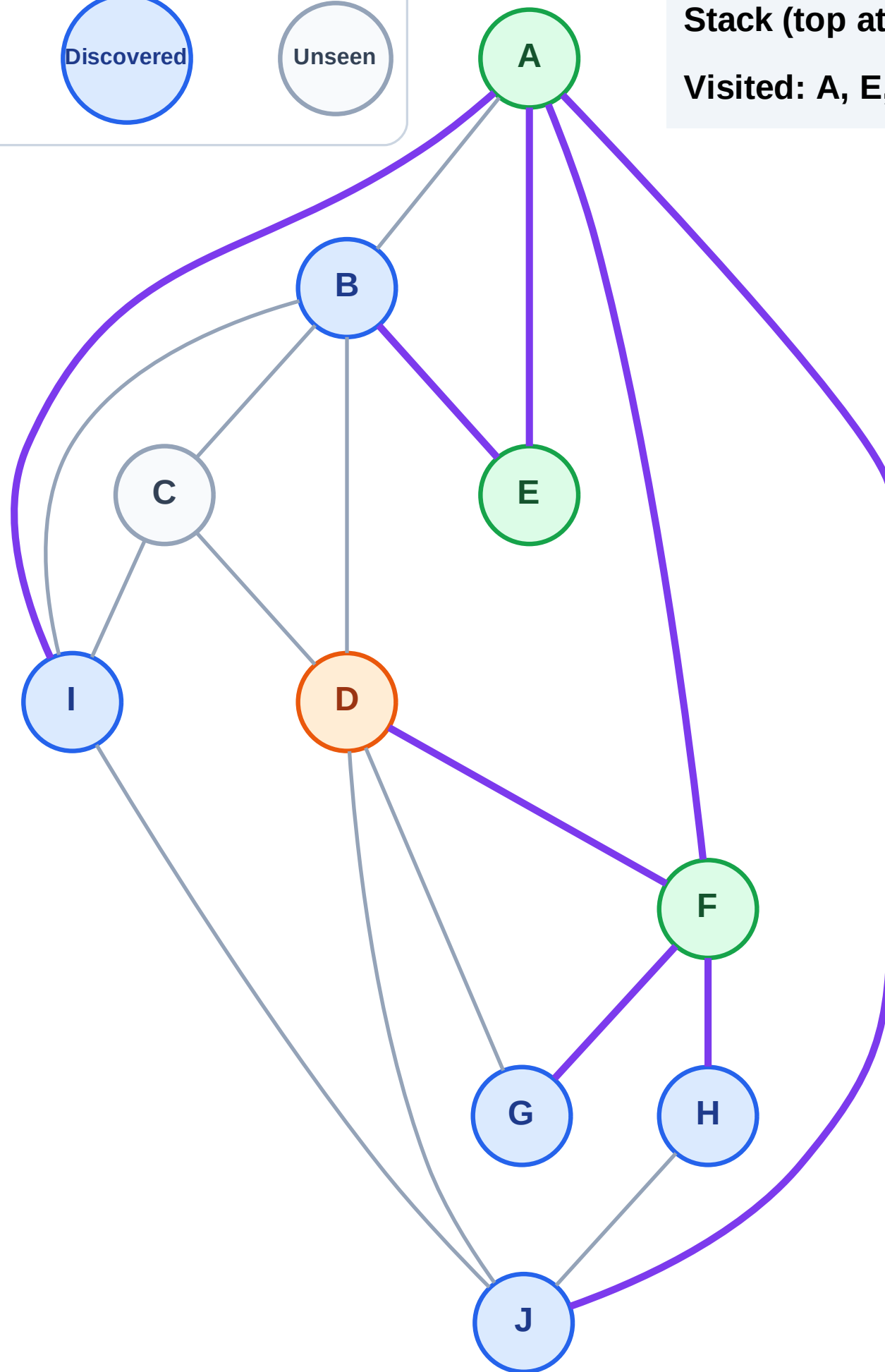
Discovered



Unseen

Stack (top at right): B | J | I | H | G

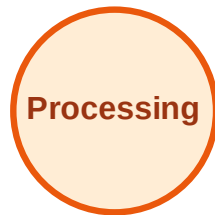
Visited: A, E, F



# DFS Traversal

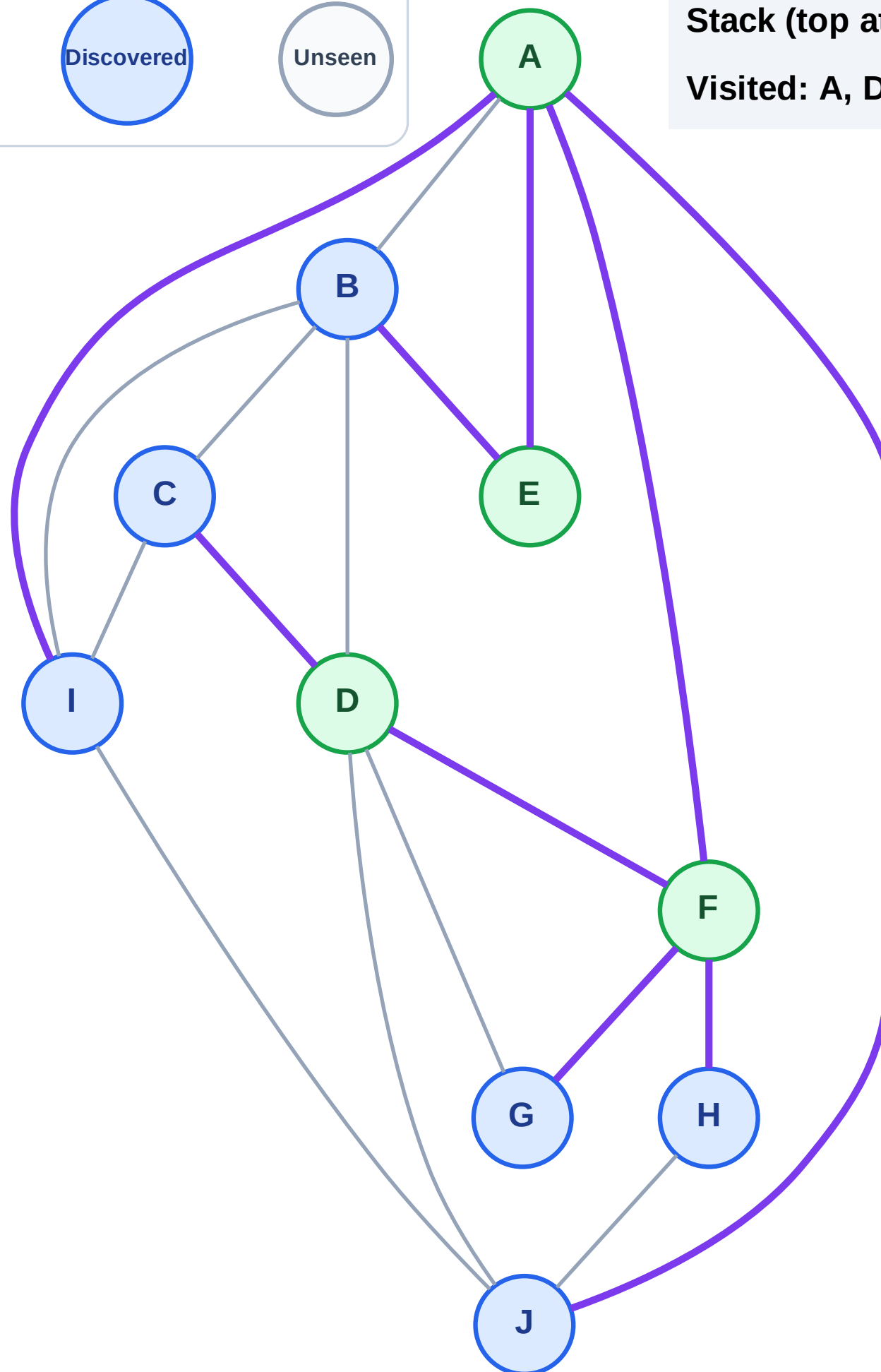
Step 9: Discover C from D

## Legend



Stack (top at right): B | J | I | H | G | C

Visited: A, D, E, F





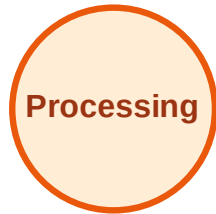
# DFS Traversal

Step 10: Process node C

## Legend



Complete



Processing



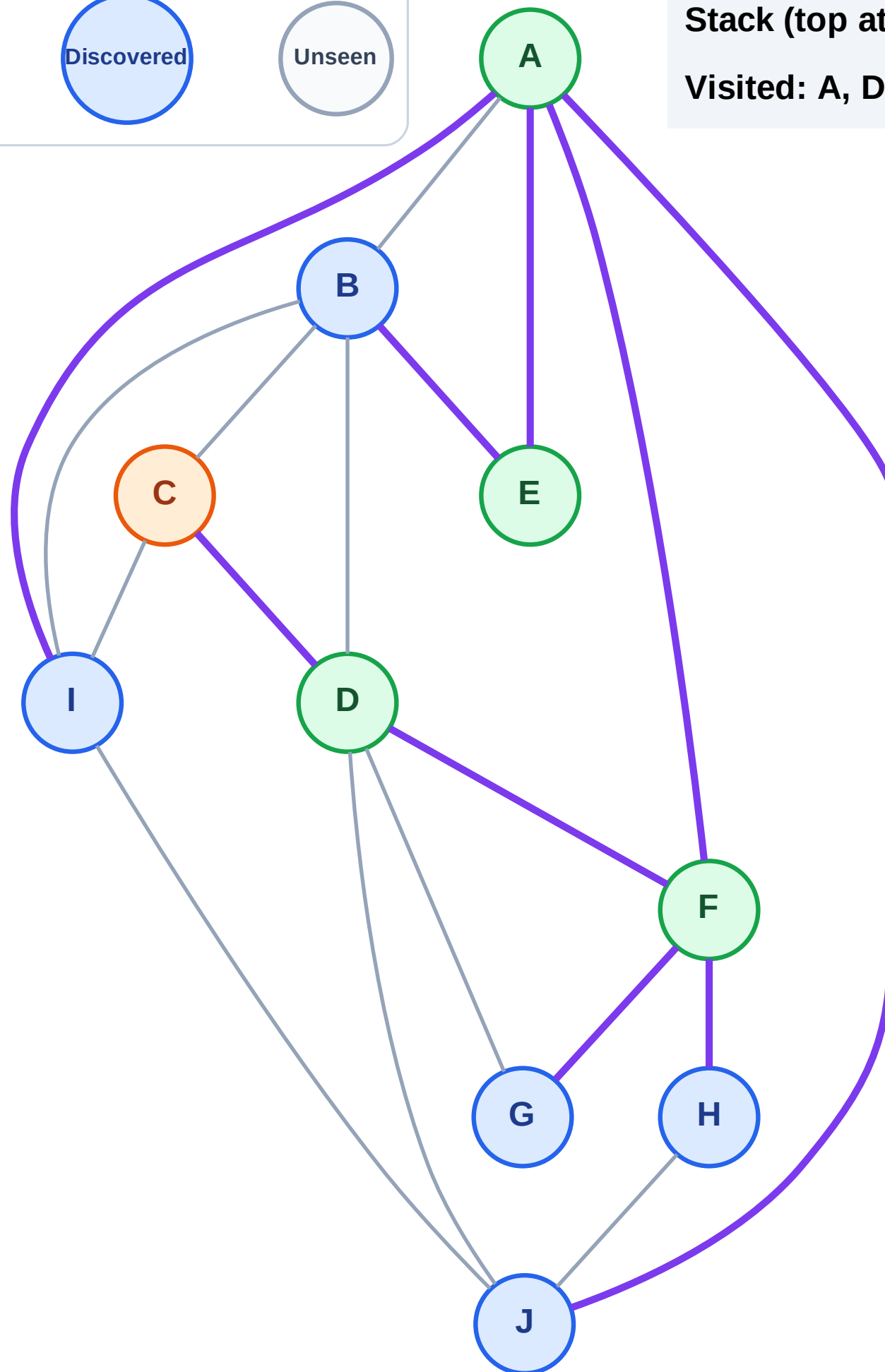
Discovered



Unseen

Stack (top at right): B | J | I | H | G

Visited: A, D, E, F



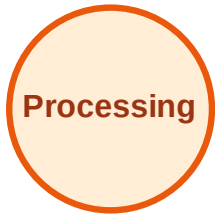
# DFS Traversal

Step 11: No new neighbors from C

## Legend



Complete



Processing



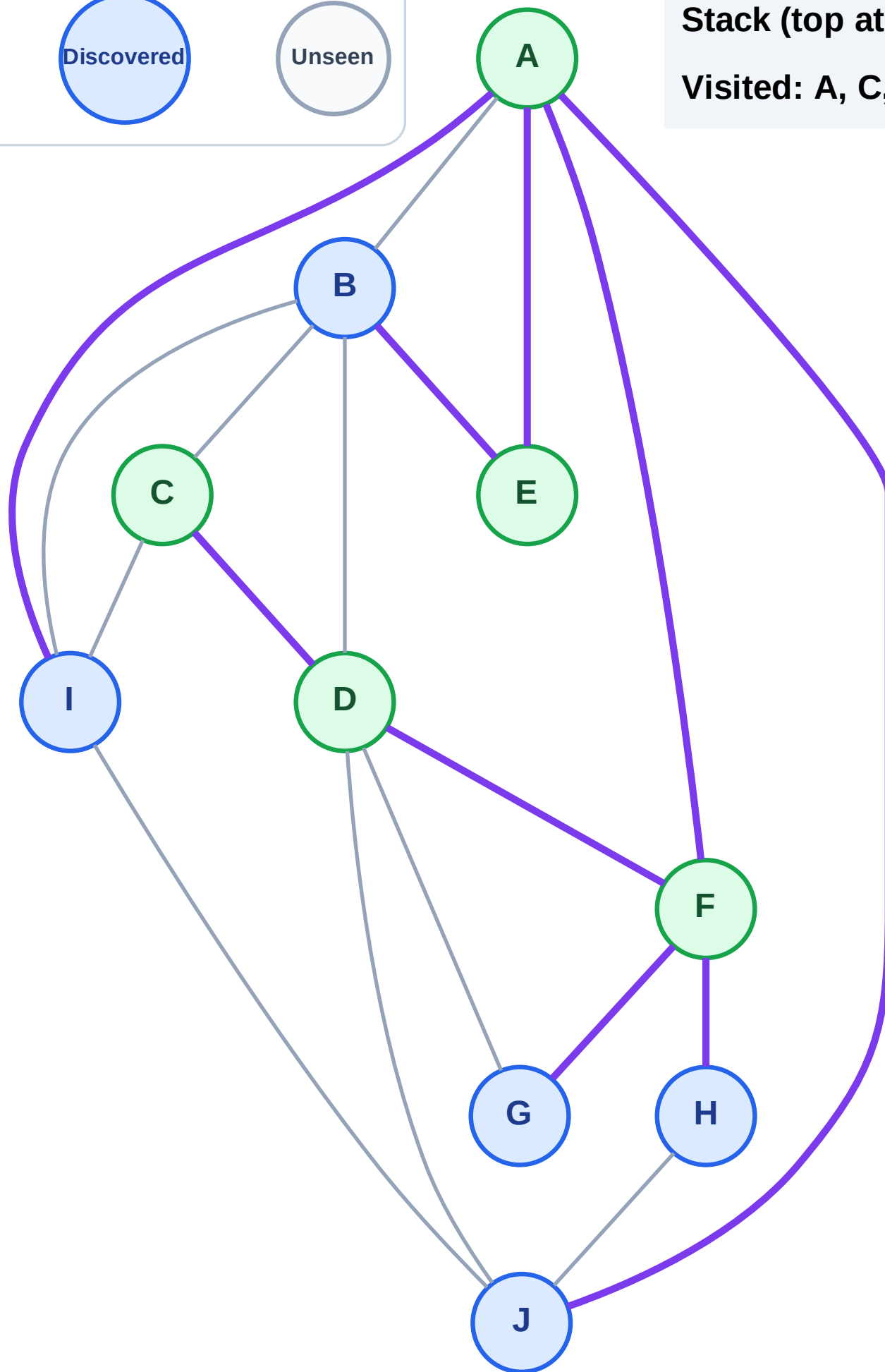
Discovered



Unseen

Stack (top at right): B | J | I | H | G

Visited: A, C, D, E, F





# DFS Traversal

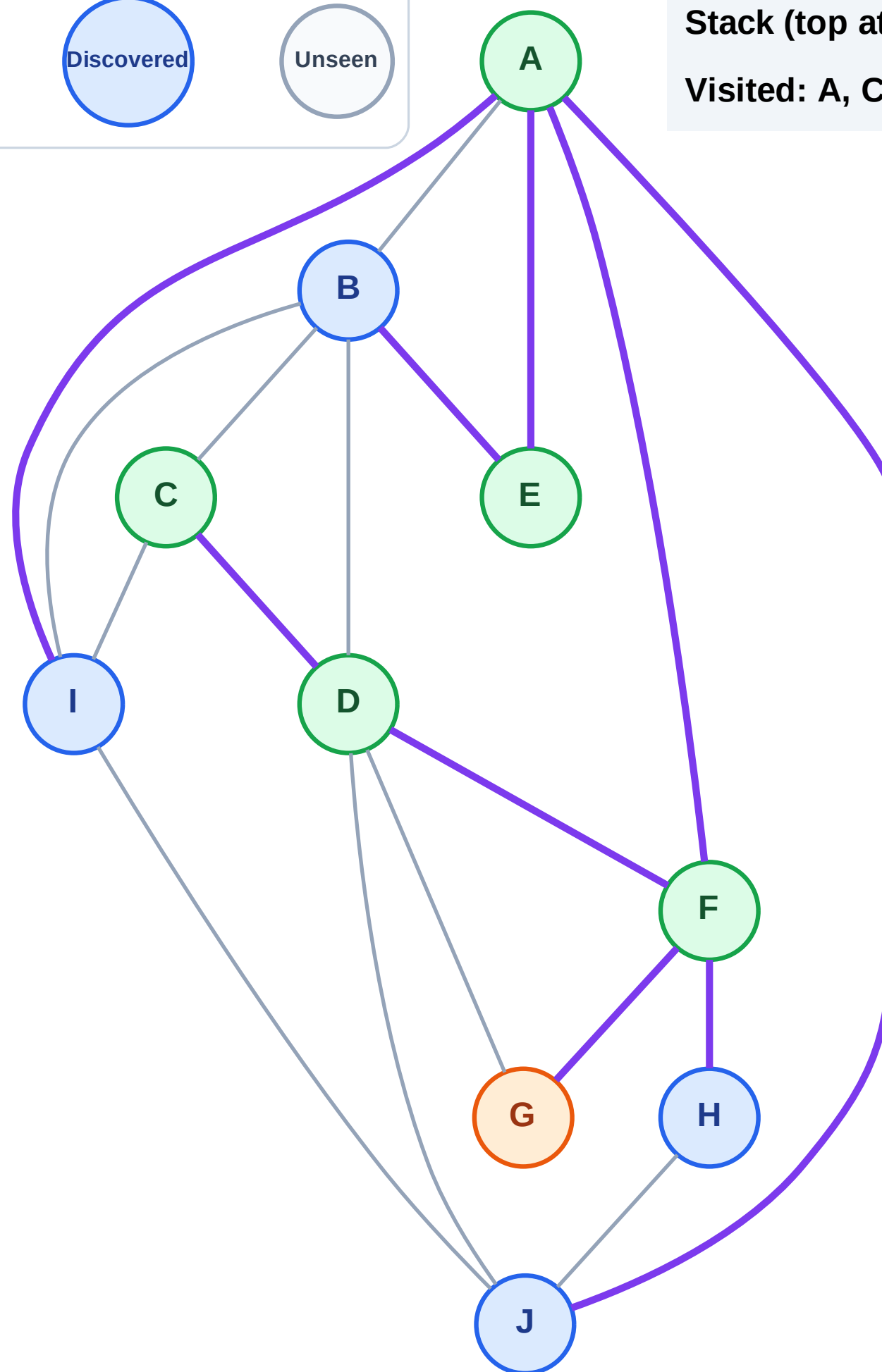
Step 12: Process node G

## Legend



Stack (top at right): B | J | I | H

Visited: A, C, D, E, F



# DFS Traversal

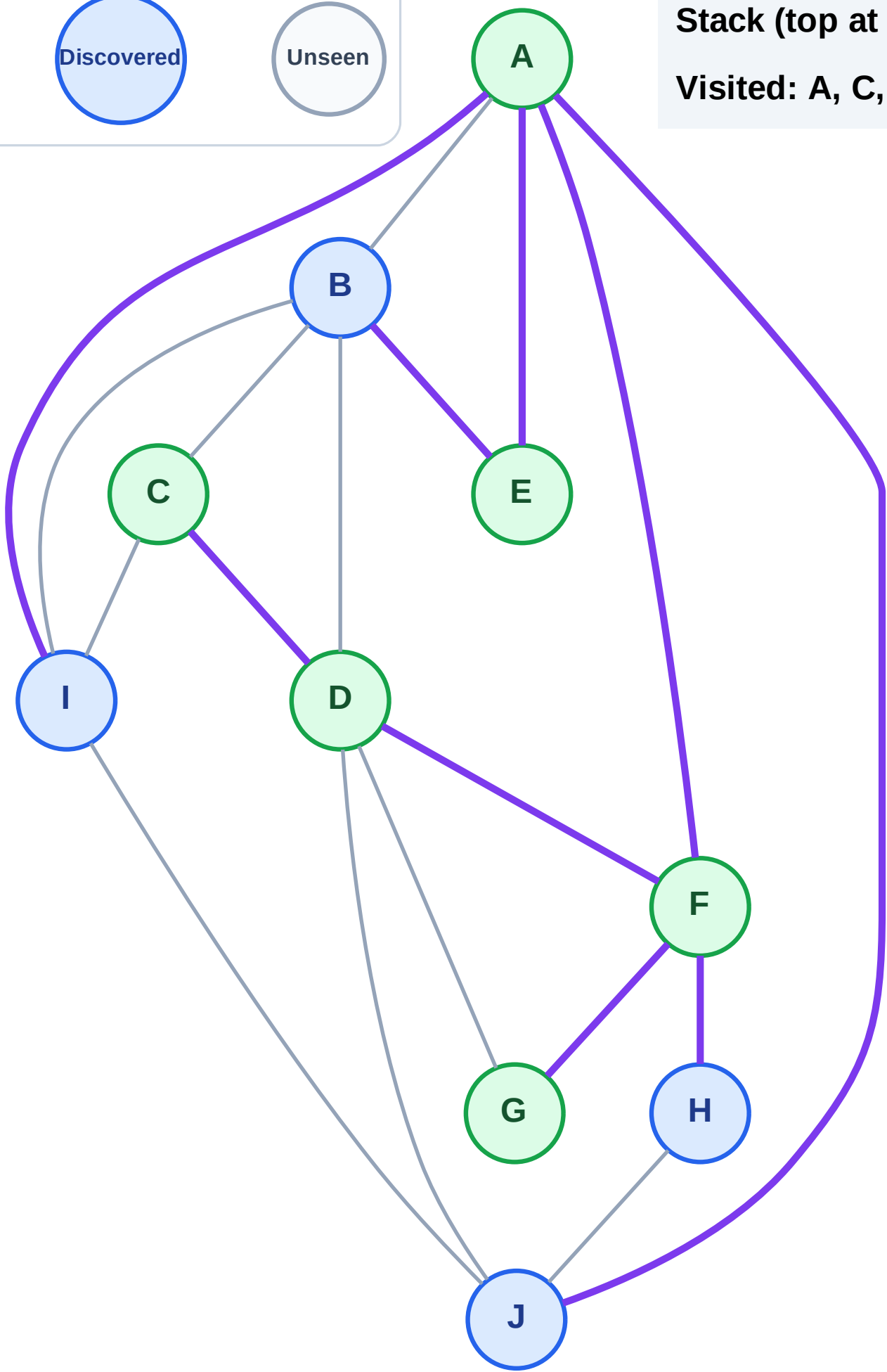
Step 13: No new neighbors from G

## Legend



Stack (top at right): B | J | I | H

Visited: A, C, D, E, F, G



# DFS Traversal

Step 14: Process node H

## Legend

Complete

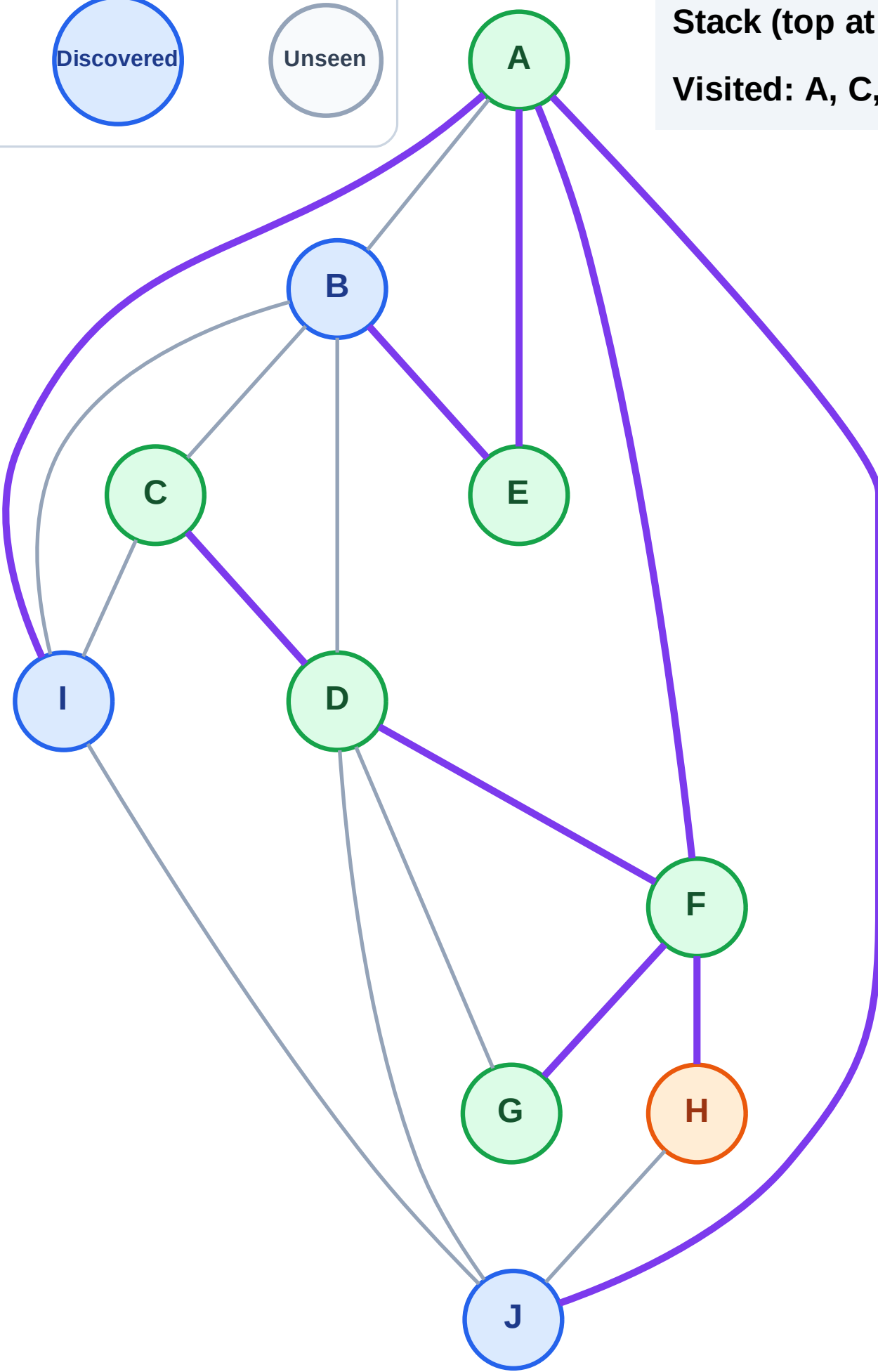
Processing

Discovered

Unseen

Stack (top at right): B | J | I

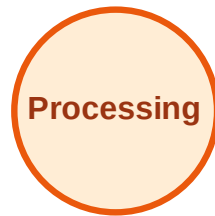
Visited: A, C, D, E, F, G



# DFS Traversal

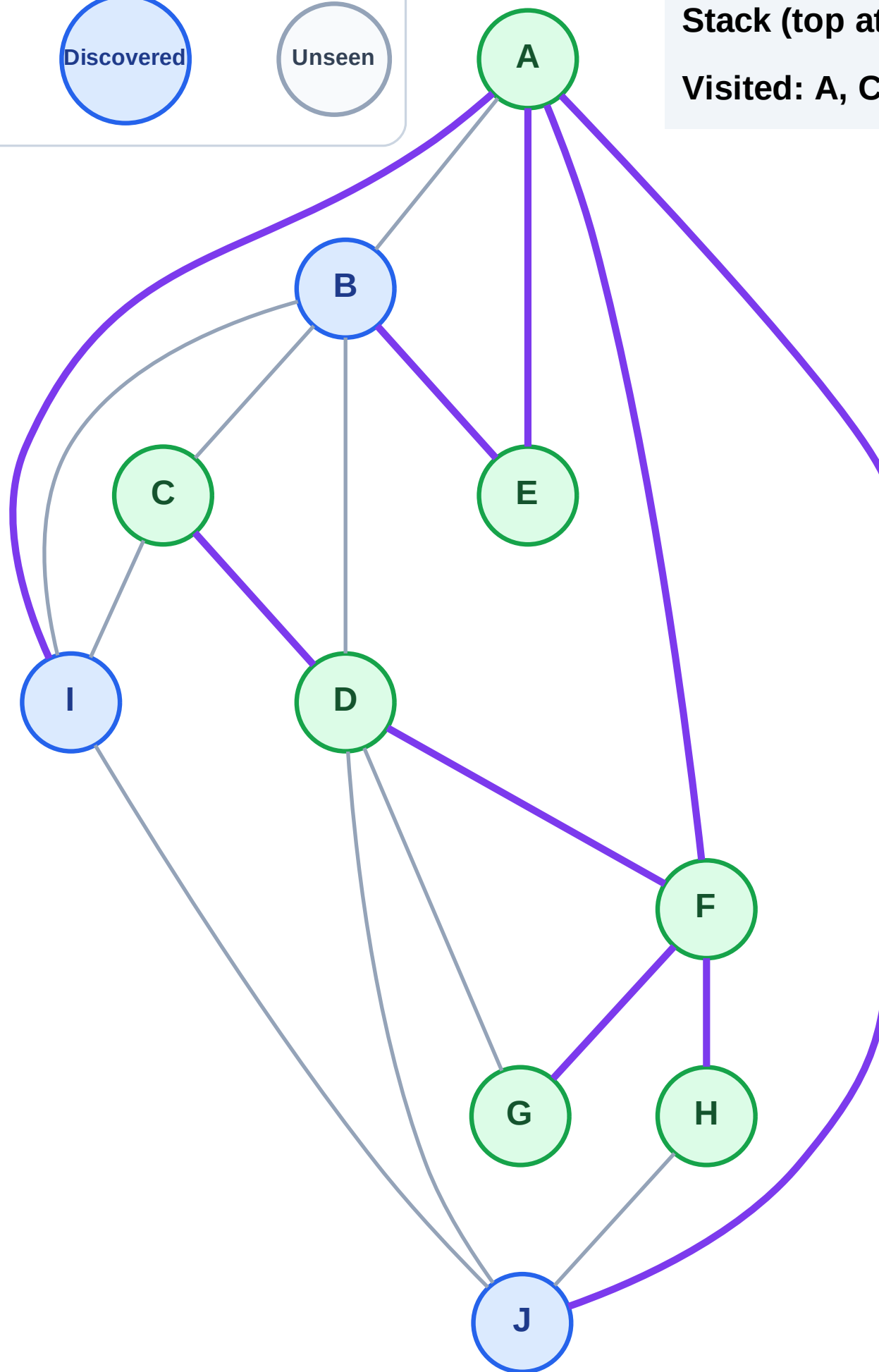
Step 15: No new neighbors from H

## Legend



Stack (top at right): B | J | I

Visited: A, C, D, E, F, G, H



# DFS Traversal

Step 16: Process node I

Legend

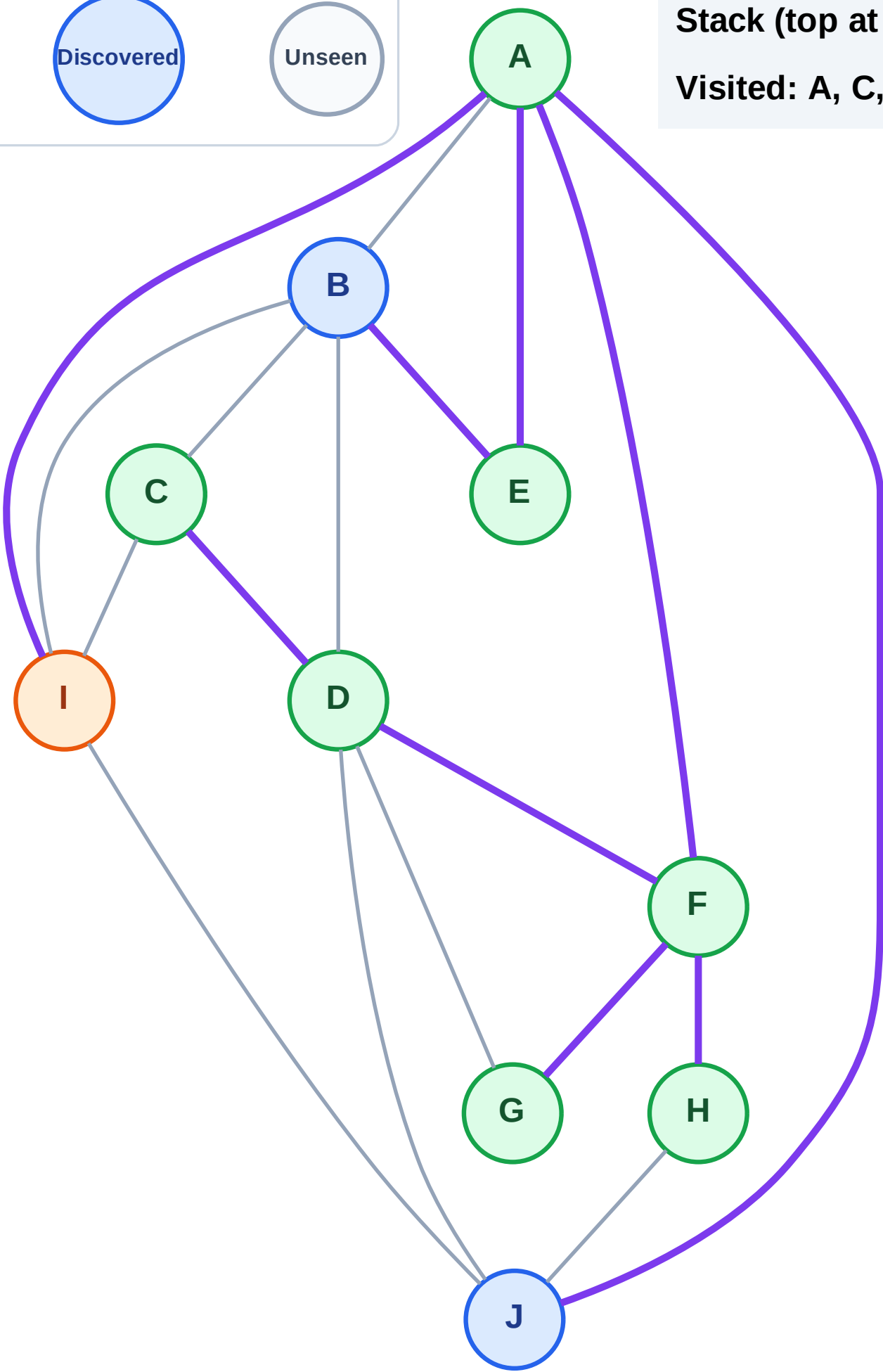
Complete

Processing

Discovered

Unseen

Stack (top at right): B | J  
Visited: A, C, D, E, F, G, H



# DFS Traversal

Step 17: No new neighbors from I

## Legend

Complete

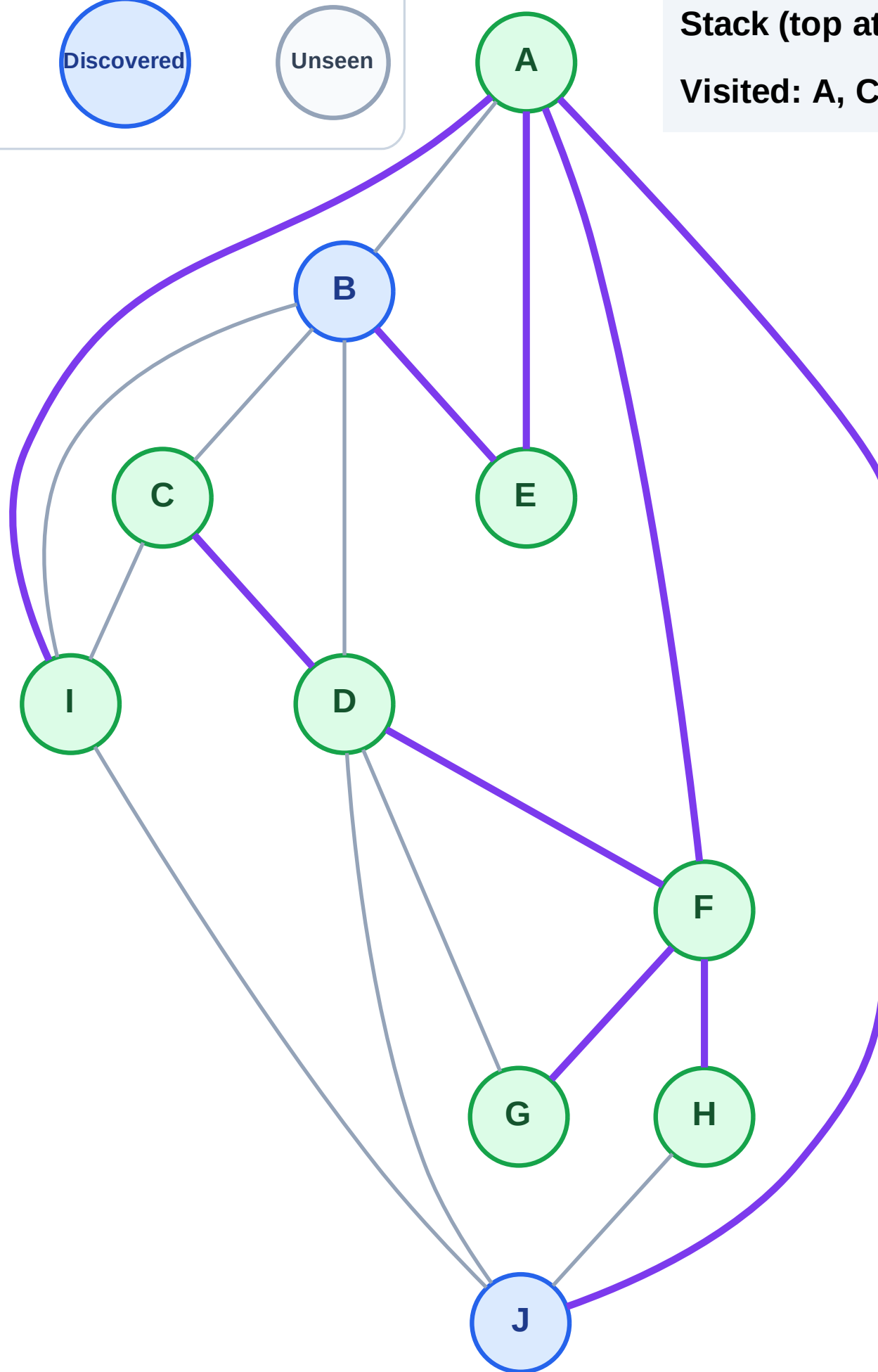
Processing

Discovered

Unseen

Stack (top at right): B | J

Visited: A, C, D, E, F, G, H, I





# DFS Traversal

Step 18: Process node J

Legend

Complete

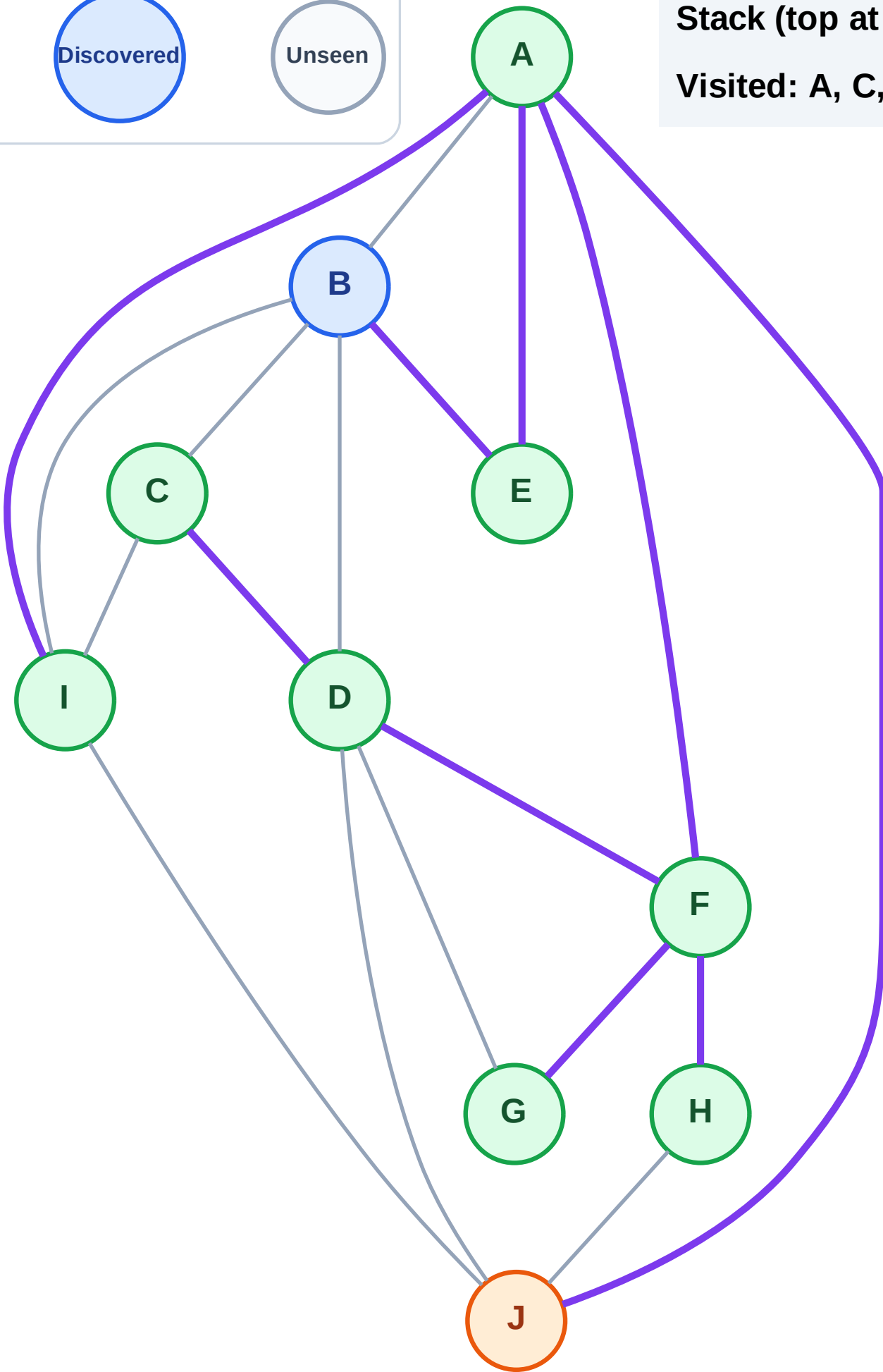
Processing

Discovered

Unseen

Stack (top at right): B

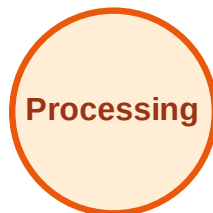
Visited: A, C, D, E, F, G, H, I



# DFS Traversal

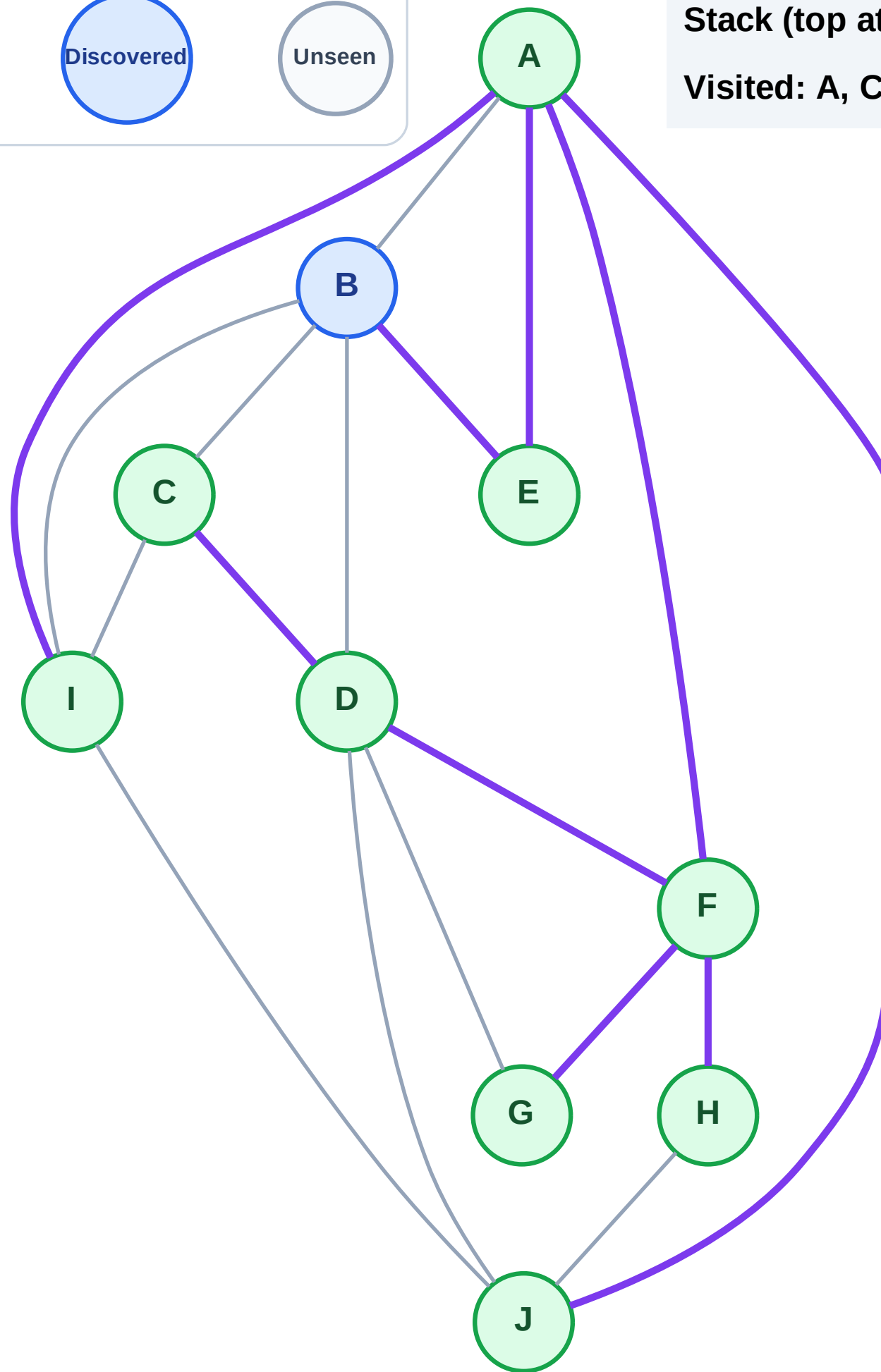
Step 19: No new neighbors from J

## Legend



Stack (top at right): B

Visited: A, C, D, E, F, G, H, I, J

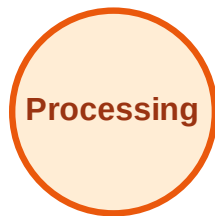




# DFS Traversal

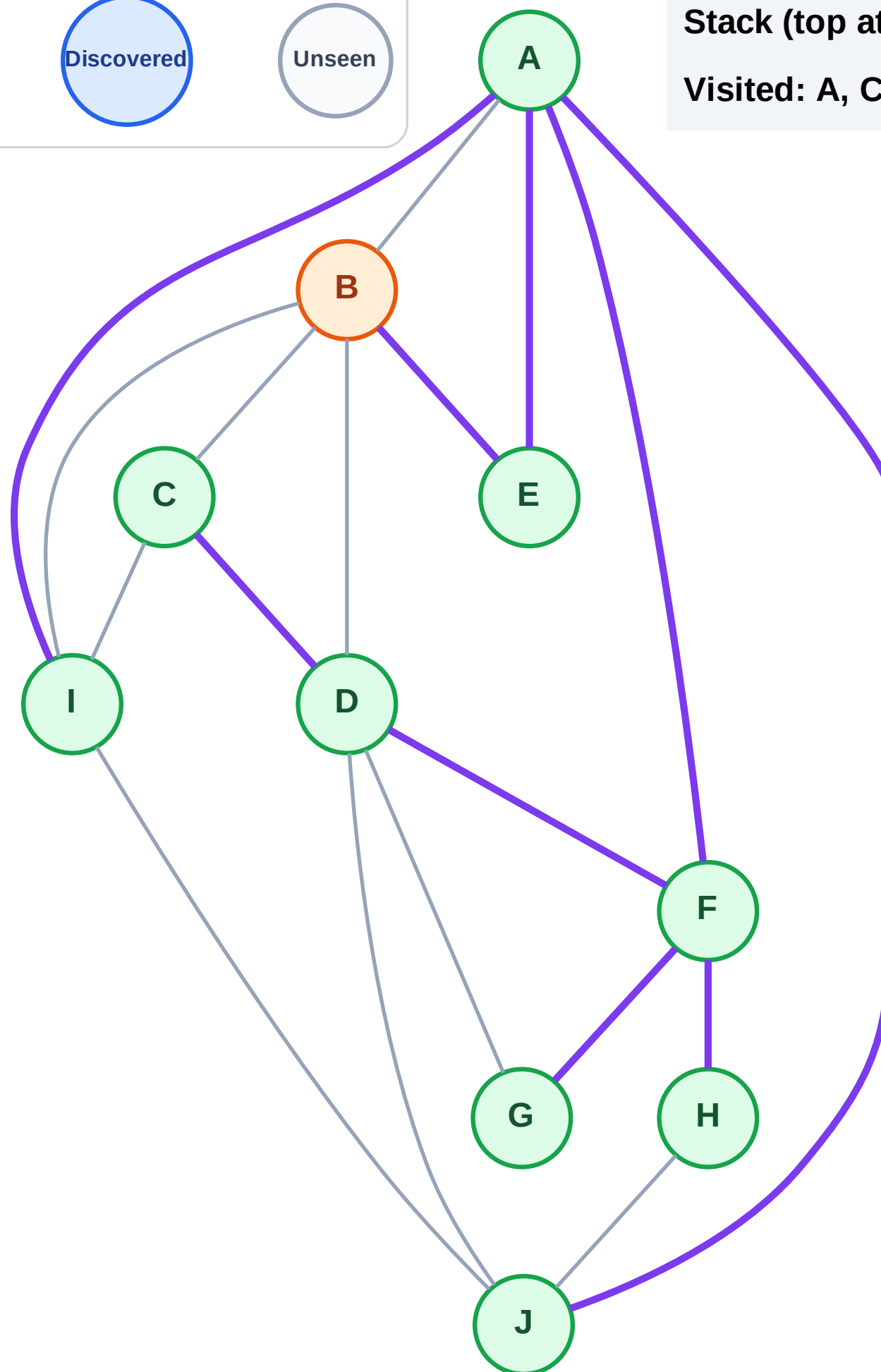
Step 20: Process node B

## Legend



Stack (top at right): (empty)

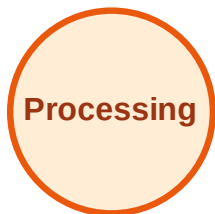
Visited: A, C, D, E, F, G, H, I, J



# DFS Traversal

Step 21: No new neighbors from B

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

